

Great Financial Crisis and Dollar Dominance

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Abstract

The majority of international trade transactions are invoiced in US Dollars, which has key implications for the global macroeconomy. Understanding the forces that sustain this equilibrium is therefore crucial. This paper studies how temporary financial shocks can generate persistent shifts in firms' invoicing decisions in international trade. Using the universe of Chilean export–import transactions from 2004–2019, we show that during the Global Financial Crisis (GFC), a temporary reversal in uncovered interest parity (UIP) between the US Dollar and the Chilean Peso led firms to substitute the US Dollar with the domestic currency when invoicing imports. This shift reduced the share of imports invoiced in US Dollars by nearly 10 percentage points, with a corresponding and persistent rise in Chilean Peso invoicing. We develop a model in which importers face complementarities between the currency used for trade invoicing and that used for financing imported inputs. UIP deviations alter borrowing costs and thus invoicing choices. Consistent with the model's predictions, we find that higher US Dollar borrowing costs increase firms' likelihood of switching to the Chilean Peso. The effect is stronger for highly dollar-exposed importers and weaker in markets where competitors predominantly invoice in US Dollars. At the aggregate level, this shift reduces the sensitivity of the terms of trade to exchange rate fluctuations by about 30%.

JEL Codes: F14, F31, F33, F41.

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1 Introduction

The global economy is dominated by the US Dollar, which serves as the dominant currency for invoicing international trade, pricing global assets, and holding reserves ([Gopinath, 2015](#); [Gopinath and Stein, 2021](#)). Existing research explains this dominance through the supply of safe assets, complementarities between financing and invoicing decisions, and the role of the US Dollar as a vehicle currency ([Farhi and Maggiori, 2018](#); [Devereux and Shi, 2013](#)). Yet, we know little about the shocks that could temporarily or permanently weaken this dominance. This paper examines how disruptions in US Dollar funding and deviations from uncovered interest parity (UIP) can produce persistent effects on firms' invoicing behavior in international trade. While past shifts in monetary dominance followed structural changes, we show that temporary shocks, such as those during the Global Financial Crisis (GFC), can also generate lasting consequences in trade invoicing patterns ([Amiti and Weinstein, 2011](#); [Ahn et al., 2011](#); [Salomao and Varela, 2022](#); [Kalemli-Özcan and Varela, 2021](#)).

Using the universe of export–import transactions from Chile between 2004 and 2019, we show that during the GFC, the deviation from UIP between the US Dollar and the Chilean Peso experienced a significant but temporary reversal in sign. This reversal induced firms to substitute the US Dollar with the domestic currency when invoicing import transactions. At the aggregate level, this shift led to a decline of just under 10 percentage points in the share of imports invoiced in US Dollar, accompanied by an almost equivalent increase of about 6 percentage points in the share invoiced in Chilean Peso.

We start by decomposing the observed change in aggregate US Dollar invoicing ([Melitz and Polanec, 2015](#)). We show that the change in invoicing is driven by incumbent importers that, after the GFC, started invoicing the same good from the same origin in domestic currency rather than in US Dollar. The switch occurs at the firm–origin–product–time level and appears permanent, as firms rarely revert once they switch.

To rationalize these empirical patterns, we build on [Bahaj and Reis \(2020\)](#). In our model, importers face within-firm complementarities between the currency used for trade invoicing and that used for financing imported inputs. Deviations from UIP affect the invoicing choice through their impact on the optimal borrowing currency. During normal times, borrowing in US Dollar is typically cheaper than in domestic currency, but during the GFC, the scarcity of dollar liquidity reversed this relationship, making peso borrowing more attractive. Firms consequently switched both borrowing and trade invoicing to domestic currency. To explain the persistence of this shift, we introduce currency- and firm-specific fixed costs following [Crowley et al. \(2020\)](#). As in [Crowley et al. \(2020\)](#), these costs decline as more firms use the same currency, capturing strategic complementarities in invoicing. However, the key innovation is the inclusion of a sunk-cost component, such as the setup cost of managing transactions in a new currency, which generates hysteresis: once incurred, the ongoing cost of using that currency remains permanently lower, making even temporary shocks yield lasting effects.

The model generates three key testable implications. First, when US Dollar borrowing cost

is higher, firms become more likely to substitute the US Dollar with the domestic currency. Second, conditional on this, firms with a higher share of import invoiced in US Dollar firms should display a greater propensity to switch. Third, this effect should weaken when a larger share of competitors invoice in US Dollar, reflecting stronger strategic complementarities in invoicing. We test these predictions using both cross-sectional variation and an event-study design. The event study shows that importers with higher pre-GFC exposure to US Dollar imports were more likely to switch to the Chilean Peso when dollar financing became relatively more expensive at the onset of the GFC. Conversely, firms more exposed to competitors invoicing in US Dollar were less likely to switch, consistent with strategic complementarities across importers. Finally, we find strong evidence of sunk costs in invoicing choices: past invoicing decisions robustly predict current ones, supporting the model’s mechanism that generates hysteresis in currency use.

To conclude, we examine the macroeconomic implications of the decline in the aggregate share of dollar-denominated imports, particularly its effects on exchange rate pass-through, the dynamics of the terms of trade, trade balance, and import price sensitivity. Consistent with previous literature, invoicing plays a crucial role in determining the degree of exchange rate pass-through, which tends to be higher when transactions are invoiced in a dominant currency and lower when invoiced in a local currency ([Gopinath et al., 2010](#); [Barbiero, 2021](#)). As a result, by 2019, the sensitivity of import inflation to exchange rate fluctuations had declined by 5% compared to 2007, while the sensitivity of the trade balance and terms of trade had fallen by approximately 30%.

Related Literature Most models of currency dominance have in common the fact that they either define under which conditions a currency becomes dominant or they define under which conditions a currency remains dominant ([Eichengreen et al., 2018](#); [Chahrour and Valchev, 2022](#); [Farhi and Maggiori, 2018](#); [Gopinath and Stein, 2021](#)).¹ Fewer papers instead focus on forces and shocks that can perturb the current dollar dominant equilibrium. [Chahrour and Valchev \(2023\)](#) and [Corsetti et al. \(2022\)](#) share the same perspective in analyzing how geo-political fragmentation and Brexit, respectively, affect the current dominant currency equilibrium. [Bahaj and Reis \(2020\)](#) and [Bahaj and Reis \(2022\)](#) focus on currency competition in the central banks’ swap line market as a channel through which the People’s Bank of China jump-started the international use of the renminbi, challenging the dominant role of the dollar. Our paper complements this literature concerned with discontinuities in the international monetary system as we shed light on how large macroeconomic events such as the Great Financial Crisis might impair the dominance of a single currency and the mechanism that could lead to the rise of another competitive currency.²

¹Other recent contributions such as [Gopinath et al. \(2020\)](#) and [Mukhin \(2022\)](#) focus on understanding the consequences that the US dollar dominance has on international trade dynamics, and the global economy in general. Refer to [Gourinchas et al. \(2019\)](#) for a survey.

²Only few other papers share the same approach and focus on a large aggregate transition in invoicing patterns like in our setting: [Garofalo et al. \(2024\)](#), which focuses on the UK after Brexit, and [Mehl and Mlikota \(2023\)](#) on the EU enlargement.

Our theoretical framework emphasizes the crucial role of trade finance. International trade transactions require external financing, which is typically heavily dollarized and sourced locally via domestic banks (Niepmann and Schmidt-Eisenlohr, 2017). The forces driving currency dominance in trade and finance are closely intertwined. The availability of safe, liquid, short-term government-backed liabilities fosters invoicing dominance, as shown by Coppola et al. (2023) and Eren and Malamud (2022). Similarly, Chahrour and Valchev (2023) highlights that dollar dominance arises from the need for both parties in an international trade transaction to post collateral. Gopinath and Stein (2021) proposes a feedback loop between invoicing and financing choices, showing that UIP deviations are necessary to the dominance of a currency. While their contribution is theoretical, we empirically document how UIP shocks can affect invoicing decisions, in line with their predictions.^{3 4}

Moreover, Kalemli-Özcan and Varela (2021) and Salomao and Varela (2022) empirically document how UIP deviations influence foreign investment and the currency of invoicing for investment financing, respectively. We complement these contributions by providing evidence that the scarcity of dollar-denominated trade finance – or, equivalently, large UIP deviations – during the GFC not only affected trade volumes (Bruno and Shin, 2019; Amiti and Weinstein, 2011), but also impacted invoicing decisions in international trade flows, with potential long-term consequences for dollar dominance.

Extensive literature focus on the firm-level determinants of invoicing choices and heterogeneous pass-through rate using micro-level data. Several papers have focused on the importance of variable markups and strategic complementarities in price-setting (Amiti et al., 2019, 2022), international input intensity and operational hedging (Amiti et al., 2014), firm characteristics such as size and productivity (Berman et al., 2012).⁵ We show that, on one hand, financial considerations are a key determinant of invoicing choices. On the other hand, standard firm-level margins are crucial not only for understanding cross-sectional differences in invoicing patterns but also for explaining the adoption of different currencies.

The paper is organized as follow. Sections 2 and 3 describe the data and invoicing dynamics in Chile, respectively. Section 4 presents the theoretical framework. Section 5 provides evidence consistent with the key theoretical mechanisms. Section 6 assesses the aggregate implication of the observed changing invoicing patterns.

³Differently from us Farhi and Maggiori (2018) explores the impact of invoicing on safe asset determination, while we focus on the same link but in the opposite direction.

⁴In our analysis, we abstract from the possibility of foreign exchange hedging (Lyonnet et al., 2022). Using data from Chile, Alfaro et al. (2021) documents the development and increased depth of the hedging market in the country starting in the early 2000s. While hedging is a relevant option for managing exchange rate risk when invoicing in foreign currency, our work is complementary in focusing on the role of dollar borrowing, which is generally cheaper on average.

⁵See Burstein and Gopinath (2014), Itskhoki (2021), and Gopinath and Itskhoki (2022) for the most recent survey articles on this vast literature.

2 Data

Customs and Invoicing Data The primary data source is the universe of Chilean trade transactions from the Chilean Customs Agency (Aduanas). The agency provides transaction-level data for each firm, product, destination/origin, currency, and day, covering both exports and imports. The dataset includes the origin or destination country, the quantity exchanged (in kilograms and the product’s unit of measure), the 8-digit HS classification of goods, and the transaction value (FOB and CIF) in US dollars. Crucially, it also records the invoicing currency, covering major currencies (US dollar, Euro, Chilean peso, UK pound, and Chinese yuan) as well as regional Latin American currencies. Additionally, an anonymous firm identifier allows for constructing a panel of Chilean importers and exporters. Import transaction data are available from 2007 to 2023, while export transaction data cover 2009 to 2023. Since our main interest is the impact of the Great Financial Crisis (GFC) on the use of the US dollar in international trade, we supplement the Chilean Customs Agency dataset with a similar dataset from [Garcia-Marin et al. \(2019\)](#).⁶ This additional dataset contains the same information as the official customs records but starts in 2004, allowing us to capture the pre-GFC period and rule out potential pre-trends. The only difference is that it covers customs transactions from Chile’s main trading partners. However, it accounts for 98% of total Chilean trade in value and quantity compared to the official data, which give us confidence in the quality and the validity of the data.

Despite the presence of multiple currencies in aggregate imports, Table 1 shows that multi-currency invoicing within a firm–origin–product–year unit (f_{opt}) is rare. Nearly 97% of f_{opt} transactions are invoiced in a single currency, and a similar pattern holds when considering the share of import value at the f_{opt} level. These statistics suggest that restricting attention to single-currency users is not a strong assumption, as it encompasses almost all observations and trade flows in our sample. Moreover, the prevalence of single-currency firms at the f_{opt} level resolves a potential identification issue that arises when firms use multiple currencies within the same f_{opt} . In that case, it would be impossible to distinguish whether a firm is switching invoicing currencies for imports from a given origin in a given year, or importing the same good from different exporters, an issue encountered, for instance, in [Corsetti et al. \(2022\)](#) using UK data.

Table 1: Number of Invoicing Currencies within Firm-Product-Origin-year Quartet

No. of Currencies	f_{pot} Quartets	Share (%)	Share of Trade (%)
1	12,437,205	96.67	93.48
2	407,506	3.17	6.13
3	19,851	0.15	0.36
4+	488	0.00	0.02

Notes: The Table reports summary statistics on multi-currency use at the firm–product–origin–year (f_{opt}) level. Specifically, we count the total number of different currencies that a firm uses to import a product (defined at the HS8 level) from a given origin in a given year.

⁶We kindly thank Santiago Justel for sharing the dataset with us.

Additional data We also use publicly available data from Chile’s financial regulatory authority, the *Comision para el Mercado Financiero* (CMF), which collects and provides data on the banking industry. They provide average monthly deposit and lending rates, distinguishing between domestic and foreign currency-denominated loans, as well as variations by maturity, loan size, and client type (commercial vs. non-commercial). Our focus is on commercial lending rates for short-term (less than one year) loans denominated in Chilean pesos and US Dollars. Short-term loan rates are particularly relevant, as trade financing for international transactions heavily depends on this type of credit (Amiti and Weinstein, 2011; Ahn et al., 2011; Schmidt-Eisenlohr, 2013). The data cover the period around the GFC but are only available until 2013, when regulatory changes disrupted consistent comparisons between domestic and US dollar-denominated loans. As a result, we can document financing cost dynamics primarily between 2007 and 2009, along with adjacent periods. Consequently, our analysis does not include the COVID-19 shock or the global financing disruptions of 2020.

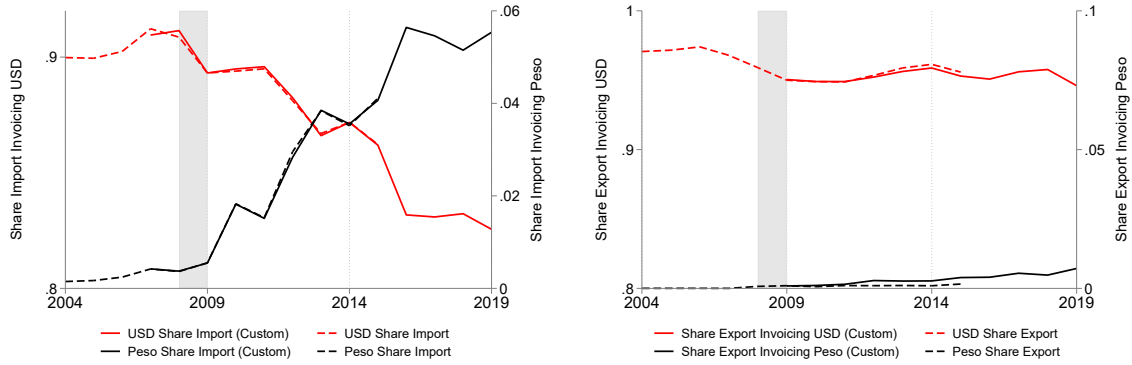
We construct time-series of uncovered interest parity deviations with respect to the US dollar by using a balanced panel of twelve currencies from January 2006 to December 2012, including the euro, Brazilian real, Canadian dollar, Swiss franc, Mexican peso, British pound, South African rand, Turkish lira, Hungarian forint, Israeli new shekel, Malaysian ringgit, and Polish zloty. Daily spot and one-month forward exchange rates are sourced from Datastream and Thomson Reuters, with all rates defined against the US dollar. To compute the one-month interest rate, we take the difference between the logarithm of the one-month forward exchange rate and the logarithm of the spot exchange rate, assuming covered interest parity holds. We then calculate weekly averages for spot exchange rates, one-month interest rate differentials, and deviations from uncovered interest parity. Lastly, we use a collection of standard macro-level variables from the Central Bank of Chile, the IMF, or the OECD.

3 Macro and Micro Dynamics in Invoicing

In this section, we show that the share of aggregate imports invoiced in US Dollars decreased by roughly 10 percentage points after the GFC, while the share invoiced in Chilean pesos increased by about 6 percentage points. Using a decomposition in the spirit of Melitz and Polanec (2015), we find that the decline in the US Dollars share is mainly driven by existing importers substituting the US Dollars with another currency when importing the same good (HS8) from the same origin in a given quarter. Next, using an event study design, we show that importers switched from US Dollars to Chilean Pesos after the GFC at the *fact* level, and that this switch appears to be permanent.

Aggregate. Following the Great Financial Crisis, the share of imports invoiced in US dollars declined by 10 p.p. while no change in invoicing patterns happened on the export side. Figure 1 shows that at the onset of the GFC, the share of imports denominated in US dollars is approximately 90%. Over a time span of 10 years, it declines until it amounts to just above

Figure 1: Aggregate Invoicing Shares for Imports and Exports



Note: Figure 1 shows the time series of the aggregate share of imports denominated in US dollars and Chilean pesos for Chile between 2004-2019. The left vertical axis of Figure 1 reports the share of Chilean imports denominated in US dollar. The right axis shows the share of Chilean imports denominated in Chilean pesos. Data from 2007 (2009) for imports (exports) are from Chilean Customs Agency. Data before 2007 (2009) are from [Garcia-Marin et al. \(2019\)](#). The grey shaded area represents the NBER recession period.

80% leading to a decline of almost 10 p.p.. In addition, the share of imports invoiced in Chilean Peso increased by a virtually 0% in 2004 to approximately 6% in 2019. Notably, we do not observe such changes in the share of exports invoiced in different currencies. In fact, the shares of exports denominated in the main currencies (US dollar, euro, and Chilean peso) remain stable over time.

Firm Level Decomposition. To understand what are the margins that drive the dynamics of the aggregate invoicing shares reported in Figure 1, we perform a dynamic Olley-Pakes decomposition in the spirit of [Melitz and Polanec \(2015\)](#). The goal of this section is to quantify the relative role played by i) firms' entry and exit, ii) imports reallocation between and within firms, and iii) firms' sourcing strategies at the origin $\times$ product level.

The change in the share of total imports invoiced in currency k at time t , $\Delta\Lambda_t^k$, can be decomposed as follow:

$$\Delta\Lambda_t^k = \underbrace{\sum_{i \in inc} \lambda_{it}^k s_{it} - \sum_{i \in inc} \lambda_{it-1}^k s_{it-1}}_{\Delta \text{Incumbents}} + \underbrace{\sum_{i \in ent} \lambda_{it}^k s_{it} - \sum_{i \in ex} \lambda_{it-1}^k s_{it-1}}_{\text{Net Entry}} \quad (1)$$

where λ_{it}^k and s_{it} are the share of firm i 's total imports invoiced in currency k at time t and the import share of firm i at time t on the total imports of Chile, respectively. Thus, $\lambda_{it}^k = \frac{\text{ToT Imp}_{it}^k}{\sum_k \text{ToT Imp}_{it}^k}$ and $s_{it} = \frac{\text{ToT Imp}_{it}}{\sum_i \text{ToT Imp}_{it}}$. A firm i is considered an incumbent (*inc*) if $s_{it} > 0$ and $s_{it-1} > 0$, entrant (*en*) if $s_{it} > 0$ and $s_{it-1} = 0$, and exiting (*ex*) if $s_{it} = 0$ and $s_{it-1} > 0$. In other words a firm is considered: an incumbent if it imports goods for two consecutive periods, an entrant firm if it starts importing in the current period and, an exiting firm if it stops importing in the current period. Clearly, a firm only belongs to one of the three groups at each point in time.

This first layer of the decomposition allows us to assess whether the decline in the share of aggregate imports invoiced in US dollars is driven by incumbent firms or by net entry. How-

ever, it does not reveal whether incumbents reduce their invoicing in US dollars because they are importing less overall, or because they continue importing the same amount but switch the invoicing currency. To see this, suppose there are no entrants or exiters so that the last two terms in Equation (1) disappear. In that case, we cannot tell whether the aggregate decline is driven by changes in market shares, s_{it} , or by changes in firm-level US Dollar invoicing shares, λ_{it}^k . To answer this question, we turn to the second layer of the decomposition.

Indeed, we further decompose Equation (1) to quantify the change in the aggregate share of incumbents that is accounted for by within-firm dynamics and between-firm reallocation:

$$\lambda_{inc,t}^k - \lambda_{inc,t-1}^k = \underbrace{\sum_{i \in inc} \Delta \tilde{s}_{it} \overline{\lambda_{i,t}^k}}_{\text{Incumbent - Between}} + \underbrace{\sum_{i \in inc} \Delta \lambda_{i,t}^k \overline{\tilde{s}_{it}}}_{\text{Incumbent - Within}}. \quad (2)$$

We introduce the following notation: $\overline{x_t} \equiv \frac{1}{2}(x_t + x_{t-1})$ and $\Delta x_t \equiv x_t - x_{t-1}$. Thus, the bar operator denotes the average of variable x between the current and preceding period, while the delta operator measures the change in x across two consecutive periods. Moreover, $\tilde{s}_{it}$ denotes the import share of firm i at time t relative to the total imports of incumbent firms. Hence, $\tilde{s}_{it} = \frac{\text{Tot Imp}_{it}}{\sum_{i \in inc} \text{Tot Imp}_{it}}$, so that $\Delta \tilde{s}_{it}$ measures the change in firm i 's market share within the set of incumbent firms, while $\overline{\tilde{s}_{it}}$ measures its average market share between periods t and $t - 1$. A similar interpretation holds for λ_{it}^k , which, as above, measures the share of imports invoiced in currency k over the total imports of the firm.

This layer of the decomposition allows us to disentangle the role played by changes in the market shares of incumbent firms from changes in their invoicing practices. In the first case, firms are simply importing less while keeping their US dollar invoicing share constant (between-firm contribution). In the second case, firms' average market shares, $\overline{\tilde{s}_{it}}$, remain relatively stable, but their share of imports invoiced in US dollars, $\Delta \lambda_{i,t}^k$, declines over time (within-firm substitution). Clearly, if the latter holds, the first term will be close to zero as $\Delta \tilde{s}_{it} \approx 0$, while the second term will be negative and drive the variation among incumbents.

One can go one step further. We decompose the within-firm component to assess whether the observed change in the US Dollar invoicing share of incumbent firms is driven by shifts in sourcing across origin $\times$ product pairs ($o \times g$), or by a reduction in quantities within a given origin $\times$ product. A product is defined at the HS8 level. Specifically, the change in firm i 's invoicing share in currency k , $\Delta \lambda_{it}^k$, can be written as:

$$\begin{aligned} \Delta \lambda_{it}^k = & \underbrace{\sum_{o \times g \in I_{it}} \Delta \lambda_{it,o \times g}^k \overline{\hat{s}_{it,o \times g}}}_{\text{Within Origin-Product}} + \underbrace{\sum_{o \times g \in I_{it}} \overline{\lambda_{it,o \times g}^k} ma \Delta \hat{s}_{it,o \times g}}_{\text{Between Origin-Product}} + \\ & + \underbrace{\sum_{o \times g \in I_{it}^+} \lambda_{it,o \times g}^k \hat{s}_{it,o \times g}}_{\text{Origin-Product Entry}} - \underbrace{\sum_{o \times g \in I_{it}^-} \lambda_{it-1,o \times g}^k \hat{s}_{it-1,o \times g}}_{\text{Origin-Product Exit}}, \end{aligned} \quad (3)$$

where I_{it} , I_{it}^- , and I_{it}^+ denote, respectively, the sets of origin-product pairs from which firm i

imports in both t and $t - 1$ (continuing origin-product), only in $t - 1$ (exiting origin-product), and only in t (entrant origin-product). The share $\hat{s}_{it,o \times g}$ is firm i 's share of imports from origin-product $o \times g$ in its total imports, $\hat{s}_{it,o \times g} = \frac{\text{ToT Imp}_{it,o \times g}^k}{\text{ToT Imp}_{it}^k}$, and it is therefore equal to zero when a firm does not import a given product $\times$ origin combination at time t independently of the currency k used. $\lambda_{it,o \times g}^k = \frac{\text{ToT Imp}_{it,o \times g}^k}{\sum_k \text{ToT Imp}_{it,o \times g}^k}$ and therefore is equal to zero only when a firm uses currency k to import at time t a origin $\times$ product combination. The delta and overbar operators follow the definitions above.

The intuition mirrors Equation (2). If firm i experiences no origin-product entry or exit (i.e., $I_{it}^+ = I_{it}^- = \emptyset$), the last two terms in (3) drop out. In that case, if the firm keeps constant the composition of its imports across origin-product pairs (so $\Delta \hat{s}_{it,o \times g} \approx 0$), the between origin-product component contributes little. By contrast, if the firm reduces the fraction of a given origin-product invoiced in currency k while holding its import share from that pair roughly constant, then $\Delta \lambda_{i,t,o \times g}^k < 0$ on average and the within origin-product component is negative, indicating a decline in the k -invoicing share at a finer level of disaggregation.

Lastly, we follow [Crowley et al. \(2024\)](#) and further decompose $\Delta \lambda_{it,o \times g}^k$ in the following way:

$$\Delta \lambda_{it,o \times g}^k = \underbrace{\lambda_{it,o \times g}^k \mathbf{1}_{\{A_{it,o \times g}^k\}}}_{\text{Currency Added}} - \underbrace{\lambda_{it-1,o \times g}^k \mathbf{1}_{\{B_{it-1,o \times g}^k\}}}_{\text{Currency Dropped}} + \underbrace{\Delta \lambda_{it,o \times g}^k \mathbf{1}_{\{C_{it,o \times g}^k\}}}_{\text{Within-Currency}}. \quad (4)$$

At the most granular level, the change in the share of imports invoiced in currency k from origin o and product g is: (i) equal to the share $\lambda_{it,o \times g}^k$ if the firm starts using currency k at time t , having not used it to import product g from origin o in the preceding period $t - 1$ ($\mathbf{1}_{\{A_{it,o \times g}^k\}}$); (ii) equal to the negative contribution $-\lambda_{it-1,o \times g}^k$ if the firm drops currency k in t after having used it in $t - 1$ ($\mathbf{1}_{\{B_{it-1,o \times g}^k\}}$); (iii) equal to the within-currency change $\Delta \lambda_{it,o \times g}^k$ if the firm continues using currency k across both periods ($\mathbf{1}_{\{C_{it,o \times g}^k\}}$) but its share may change over time.

Table 2: Dynamic Olley-Pakes Decomposition with Entry and Exit: Imports at Origin-Product

	US Dollar	Chilean Peso	Euro	Others
Aggregate Invoicing Share	-9.452	6.093	3.611	-0.251
Net Entry	-1.769 (18.7%)	0.778 (12.8%)	0.813 (22.5%)	0.178 (29.3%)
Incumbents	-7.683 (81.3%)	5.315 (87.2%)	2.798 (77.5%)	-0.430 (70.7%)
Incumbents	-7.683	5.315	2.798	-0.430
Within Firm	-4.560 (59.4%)	6.367 (85.8%)	-1.337 (24.4%)	-0.470 (92.0%)
Between Firm	-3.123 (40.6%)	-1.052 (14.2%)	4.134 (75.6%)	0.041 (8.0%)
Within Firm	-4.560	6.367	-1.337	-0.470
Net Entry Origin-Product	0.975 (15.0%)	2.131 (33.5%)	-2.980 (55.7%)	-0.126 (17.2%)
Within Origin-Product	-5.139 (78.9%)	3.607 (56.6%)	2.007 (37.5%)	-0.475 (65.0%)
Between Origin-Product	-0.396 (6.1%)	0.630 (9.9%)	-0.363 (6.8%)	0.130 (17.8%)
Within Origin-Product	-5.139	3.607	2.007	-0.475
Currency Net Entry	-2.170 (42.2%)	2.060 (57.1%)	1.903 (94.8%)	-0.505 (94.2%)
Within Currency	-2.968 (57.8%)	1.546 (42.9%)	0.104 (5.2%)	0.031 (5.8%)

Notes: Table 2 reports the results from the dynamic Olley-Pakes decomposition with entry and exit at the origin-product level for Chilean imports between 2007 and 2020. It corresponds to the decompositions in Equations (1), (2), (3), and (4). Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year.

Table 2 presents the results of the decomposition exercise for Chilean imports between 2007 and 2019 across four currency categories: US Dollar, Chilean Peso, Euro, and “Other” (grouping all remaining currencies). The first row reports the cumulative change in invoicing shares over the sample period. Consistent with Figure 1, the share invoiced in US Dollar fell by just under 9.45 p.p., while the Chilean Peso share increased by about 6 p.p. The Euro’s share also rose, by roughly half as much as the Peso, while “Other” currencies remained broadly stable with a slight decline.⁷

According to Equation (1), more than 80% of the cumulative decline in the US Dollar share is explained by incumbent firms ($-7.683 / -9.452 \approx 0.81$), with the remainder due to net exit. Equation (2) shows that almost 60% of the incumbents’ contribution ($-4.560 / -7.683 \approx 0.59$) stems from the within-firm component, with the rest due to between-firm reallocation. This implies that firms are not primarily reducing their import quantities, but rather continuing to import similar amounts while invoicing less often in US Dollars.

Equation (3) zooms in further to the origin-product level. The results show that the bulk of the within-firm effect arises within continuing origin-product pairs: firms keep importing the same combinations but invoice a smaller share in US Dollars over time. Finally, the bottom panel of Table 2 shows that both net currency switching and within-currency adjustments contribute roughly equally to the decline in the within origin-product component. That is, firms both drop the US Dollar in favor of alternative currencies (Currency Dropped) and, conditional on continuing to use the US Dollar, reduce the share invoiced in it (Within-Currency).

A symmetric interpretation applies for the Chilean Peso. The increase in its invoicing share is mainly driven by domestic firms expanding its use along both the intensive and extensive margins: they increasingly invoice existing origin-product imports in Chilean Pesos (positive

⁷See also Table 11 in Appendix A.

Within-Currency), and when adopting new currency in an already existing origin–product pairs, they are more likely to use the domestic currency (Currency Added).

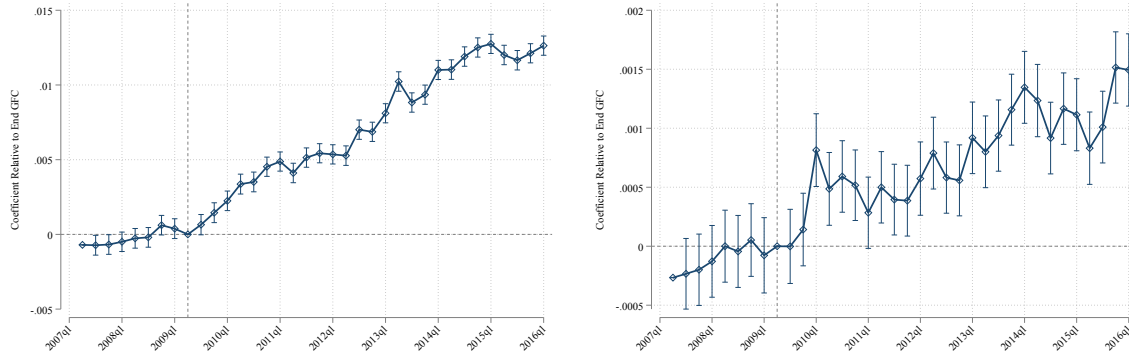
Firms Switch Permanently From US Dollar to Chilean Peso. Here, we show evidence that after the GFC, importers began invoicing transactions at the f_{opt} level in Chilean pesos instead of U.S. dollars, and that this switch appears to be permanent. To show this, we turn to the specification in Equation 5:

$$y_{f_{opt}} = \sum_{s=2007Q1}^{2016Q4} \beta_s \cdot \mathbf{1}[t = s] + \gamma_{f_{op}} + \epsilon_{f_{opt}}, \quad y_{f_{opt}} \in \{\text{Pr(PESO)}_{f_{pot}}, \text{Pr(Switch)}_{f_{pot}}\} \quad (5)$$

We run this specification using two different dependent variables. In one case, we estimate a linear probability model where the dependent variable is a dummy equal to one when the transaction at the f_{opt} level is invoiced in Chilean Peso, and zero when invoiced in US Dollar. In the second case, the dependent variable is a dummy equal to one if the firm invoiced a transaction at the f_{op} level in US Dollar at time $t - 1$ and then invoiced the same transaction in Chilean Peso at time t . This allows us to estimate whether there is an increase in the probability of switching from US Dollar to Chilean Peso before and after the GFC.⁸ The main explanatory variable is a set of quarter-year fixed effects covering the period from 2007Q1 to 2016Q4. This specification also allows us to test for the presence of any pre-trends. Consistent with the decomposition presented above, we include $f \times o \times p$ fixed effects in our preferred specification. The left panel of Figure 2 shows the estimated coefficients when the dependent variable is defined as in Equation (9), while the right panel corresponds to the case of switchers. In both cases, the coefficients are normalized with respect to the coefficient at the end of the GFC, that is 2009Q2.

⁸The number of observations used in in this case is lower because the sample is restricted to firms using a single currency at each point in time at the f_{opt} level. In addition, we focus only on firms that used US Dollar in the preceding period and in the subsequent period either the US Dollar or the Chilean Peso. In light of Table 5, this restriction corresponds to focusing only on the observations contained in positions (2,2) and (2,4).

Figure 2: Effect of GFC on Probability of Invoicing in Chilean Peso over Time



Note: Figure 2 plots the estimated coefficients from Equation (5). Each point represents the effect of a given quarter on the probability of invoicing in Chilean Peso, relative to 2009Q2 (the end of the GFC). The left panel reports results for the full sample, while the right panel restricts the sample to switching firms. The vertical dashed line corresponds to the NBER recession period for the end of the GFC. Standard errors are clustered at the firm–product–origin level.

Both panels show the absence of pre-trends in firms’ invoicing decisions before the GFC. After the GFC, the probability of invoicing in Chilean Peso relative to the US Dollar increases steadily, reaching 1.2 percentage points above the pre-GFC level by the last quarter of 2016. A similar upward trajectory is observed for the subset of switching firms, where the probability of switching from the US Dollar to the Chilean Peso is just below 0.2 percentage points higher at the end of 2016 than before the GFC.

Robustness: Alternative Specifications & Additional Results. In addition, we estimate whether the likelihood of invoicing in Chilean Peso relative to the US Dollar has increased after the GFC. To do so, we estimate a linear probability model where the dependent variable is a dummy equal to one when the transaction at the *f_{opt}* level is invoiced in Chilean Peso, and zero when invoiced in US Dollar. The main explanatory variable is a dummy identifying the period before and after the GFC, taking value one for the years 2009–2019 and zero for the years 2003–2008. Results are reported in Table 12 and we find similar results than those reported in Figure 2.⁹ Moreover, we examine whether firms that switch from US Dollar to Chilean Peso for the first time maintain this choice over time. Figure 15 shows that, one year after the initial switch, roughly 80% of *f_{op}*-level transactions continue to be invoiced in Chilean Peso. This indicates that the firms’ choice is highly persistent over time.

Robustness: Decomposing Invoicing Shares Over Time. Table 10 and Table 11 in Appendix A plot the time–series contributions between 2007 and 2020 for each layer of the decomposition reported in Equations (1), (2), (3), and (4), for imports invoiced in Chilean Peso and US Dollar, respectively. These figures are informative because they show that the

⁹The specification in this case is:

$$y_{f_{opt}} = \beta_1 \cdot \text{PostGFC}_t + \gamma_{f_{po}} + \epsilon_{f_{pot}}, \quad y_{f_{opt}} \in \{\text{Pr(PESO)}_{f_{pot}}, \text{Pr(Switch)}_{f_{pot}}\}$$

decline (increase) in the share of imports invoiced in US Dollar (Chilean Peso) is not concentrated in a particular year but instead evolves smoothly over time following the GFC. The conclusions drawn in the main section are robust: they apply not only to the cumulative decomposition reported above, but also to the time-series contributions of each layer at every point in time.

Moreover, Table 10 in Appendix A presents results from a dynamic Olley–Pakes decomposition with entry and exit at the origin level, abstracting from the product dimension. These results are qualitatively consistent with our benchmark decomposition. Finally, Table 11 in Appendix A applies the benchmark decomposition to exports, revealing no significant changes at the aggregate level.

Robustness: Origin & Product Heterogeneity. In Appendix A, we also estimate Equation (5) separately for the top twelve origins in the sample (ranked by number of observations at the f_{opt} level between 2007Q1 and 2019Q4). We repeat the exercise for the top twelve HS2 codes using the same criterion. The results, reported in Figures 13 and 14, respectively, show a similar post-GFC increase in the probability of invoicing in Chilean Peso. This confirms that the average effect reported in the left panel of Figure 2 is not driven by a specific origin but is fairly homogeneous across origins. The same conclusion holds for products imported by Chilean firms.

Taking an even more aggregate perspective, Figure 12 shows that invoicing dynamics are not driven by specific product types. Splitting the sample into three broad BEC categories, consumption goods, capital goods, and intermediate goods, we find that the share of imports invoiced in Chilean Peso increases across all three. A similar pattern emerges when we split trading partners into the United States, Europe, China, Central–South America, Asia–Middle East, Africa, and Others.

4 A Model of Currency Choice

In this section, we present the theoretical framework used to interpret the empirical invoicing patterns observed among Chilean importers. The model combines insights from the literature on invoicing choices in international trade with those on the currency denomination of firms’ borrowing costs. In particular, it builds on Bahaj and Reis (2020) to capture within-firm complementarities between the currency used to invoice international trade transactions and the currency used to denominate borrowing costs.

The framework also incorporates strategic complementarities in invoicing across importers, modeled through currency-specific fixed costs that decline with the number of firms invoicing in a given currency (Amiti et al., 2022; Crowley et al., 2020). Finally, we augment the existing setup by introducing sunk costs associated with adopting a new invoicing currency. These sunk costs generate hysteresis in invoicing decisions following temporary shocks to firms’ financing costs, allowing us to account for the persistence of the invoicing patterns

documented in Section 3.

4.1 Environment

A small open economy has a continuum of importers indexed by $i \in [0, 1]$. The firm sells a differentiated output domestically, pricing it in domestic currency. The demand faced by firm i in the domestic market d is $y_i^d = \left(\frac{p_i^d}{q^d}\right)^{-\theta}$, where q^d is a market specific demand shifter, and $\theta > 1$ is the constant demand elasticity.

Each importer combines domestic labor, l_i , and imported inputs, x_i , using a CRS Cobb-Douglas production function:

$$y_i = A_i l_i^{1-\alpha_i} x_i^{\alpha_i}, \quad (6)$$

where y_i denotes the output of firm i , α_i the share of imported inputs used in production by firm i , and $A_i \equiv \left(\frac{1}{\alpha_i}\right)^{\alpha_i} \left(\frac{1}{1-\alpha_i}\right)^{1-\alpha_i}$ is a normalizing constant. Importantly, we assume that the imported inputs, x_i , are working capital and must be paid ahead of production, while labor can be paid later when revenues are generated.

In each period, there are two sub-periods: a morning and an evening. In the morning, the firm finances the imported inputs x_i required as working capital and chooses the currency of borrowing. It can borrow either in the local currency d or in the vehicle currency v . Prices are set in the morning, are nominally sticky, and remain fixed throughout the period. In the evening, the firm purchases inputs, meets demand at the preset price, collects revenues, and repays its loans. The key feature is that the working capital choice made in the morning affects production in the evening because the bilateral exchange rate between the domestic currency and the vehicle currency, s_v , is not known at the time of the input decision. As a result, different realizations of the exchange rate alter the realized cost of production, which can generate deviations from the profit-maximizing price.

Firms face the following production technology for imported inputs x_i :

$$x_i = \min \left(\frac{x_i^d}{\eta_i}, \frac{x_i^v}{1 - \eta_i} \right). \quad (7)$$

By choosing η_i , the firm determines the relative shares of two imported inputs: x_i^d , denominated in domestic currency d , and x_i^v , denominated in vehicle currency v . In the corner solutions, if $\eta_i^* = 0$ the firm exclusively invoices inputs in vehicle currency, while if $\eta_i^* = 1$ it invoices entirely in domestic currency.

The firm must borrow to finance imported inputs and can choose between domestic and foreign currency borrowing. In principle, a firm could finance its imported inputs in a currency different from the one used to invoice international trade transactions. Following [Bahaj and Reis \(2020\)](#), however, we restrict attention to the case in which the financing currency coincides with the invoicing currency of imported inputs. This simplification is without loss of generality, as it arises optimally: firms prefer to align financing and invoicing currencies to minimize the exchange rate risk created by currency mismatches. Hence, in the model

the currency chosen to invoice borrowing costs is equivalent to the currency used to invoice imported inputs.

The firm faces two financing options: (i) borrow one unit of domestic currency and repay $(1 + i_d)$, or (ii) borrow one unit of vehicle currency and repay $(1 + i_v)$. While interest rates are known at the time of borrowing, the exchange rate between the domestic and vehicle currencies, s_v , remains uncertain.

Given the cost structure described above, the *ex post* marginal cost of production for firm i , expressed in domestic currency, is:

$$C_i(s^v, \eta_i) = \frac{1}{A_i} \left[\frac{\eta_i \times \rho_d(1 + i_d) + (1 - \eta_i) \times \rho_v s_v(1 + i_v)}{\alpha_i} \right]^{\alpha_i} \left[\frac{w}{1 - \alpha_i} \right]^{1 - \alpha_i}, \quad (8)$$

where w denotes the domestic wage rate (homogeneous across firms), and ρ_d and ρ_v are the prices of domestic and imported intermediate inputs expressed in domestic and vehicle currency, respectively. All prices and borrowing costs are taken as exogenous. Crucially, it follows from standard theory that firms optimally choose corner solutions for invoicing: imported inputs are invoiced entirely in either domestic or vehicle currency, that is $\eta_i^* \in \{0, 1\}$.

Lastly, we turn to the final key cost component: currency- and firm-specific fixed costs. Firms invoicing international transactions in currency j incur a fixed cost F_i^j . Following [Crowley et al. \(2020\)](#), we assume that this cost decreases with the number of firms using currency j . We extend their specification by introducing sunk costs in the adoption of new currencies, implying that a firm's fixed cost of using currency j depends on its invoicing choice in the previous period.

Formally, the reduced-form representation of firm i 's fixed cost of invoicing in currency j at time t is:

$$F_{it}^j = f_{it}^j - \gamma^j \omega_{t-1}^j(N_{-it}), \quad (9)$$

where $\omega_{t-1}^j(N_{-it})$ denotes the share of importers using currency j in period $t - 1$. The firm-specific component is given by

$$f_{it}^j = \begin{cases} \kappa_0^j & \text{if firm } i \text{ uses currency } j \text{ for the first time,} \\ \kappa_1^j & \text{if firm } i \text{ has previously used currency } j, \end{cases}$$

with $\kappa_1^j < \kappa_0^j$. The difference $\kappa_0^j - \kappa_1^j$ represents the sunk entry cost of invoicing in a new currency, capturing one-time managerial costs such as setting up foreign-currency bank accounts.¹⁰

The second term in Equation (9) captures externalities in adoption: the fixed cost decreases as more firms choose currency j , reflecting strategic complementarities in invoicing ([Amiti et al., 2019](#); [Crowley et al., 2020](#)).¹¹ The strength of these complementarities is gov-

¹⁰We assume that the firm-specific component depends only on the previous period's currency choice. Richer specifications could allow a longer invoicing history to matter, leading to gradual cost accumulation. See [Alessandria et al. \(2021\)](#) for a review of sunk costs in trade.

¹¹In our framework, complementarities operate across firms. Empirical evidence in Section 3 shows that

erned by γ^j , which satisfies the stability condition $0 < \gamma^j < f_{it}^j < 1$. The core idea of complementarities is that usage rises as overall utilization increases. One practical rationale is that firms may prefer to invoice in the same currency as their competitors to maintain similar exchange rate pass-through (ERPT) to import prices.

4.2 Working Capital Invoicing Choice

In this section, we solve for both the optimal invoicing decision for imported inputs and the optimal domestic pricing decision. Crucially, we obtain comparative statics results that we will later test directly in the data.

Optimal Invoicing Decision Firms are assumed to be risk-neutral and aim to maximize expected profits. They form expectations about the future realization of the exchange rate between the domestic currency and the vehicle currency, s_v . Fluctuations in exchange rate realizations affect the ex-post cost of credit in the vehicle currency, generating deviations from the optimal markup over marginal costs and reducing profits.

Given the isoelastic demand function and the imported-input technology in Equation (7), firm i maximizes ex-ante profits:¹²

$$\max_{\eta_i, p_i^d} \mathbb{E} [\pi_i^j] = p_i^d (p_i^d)^{-\theta} - \mathbb{E} [C_i(s_v, \eta_i)] (p_i^d)^{-\theta} - \eta_i F_i^d - (1 - \eta_i) F_i^v. \quad (10)$$

For a given η_i , the optimal domestic price is $p_i^{d*} = \frac{\theta}{\theta-1} \mathbb{E} [C_i(s_v, \eta_i)]$. Given p_i^{d*} , firm i chooses to invoice imported inputs in the vehicle currency ($\eta_i^* = 0$) if the following condition holds:

$$\eta_i^* = 0 \quad \text{iff} \quad \mathbb{E} [(\rho_v(1 + i_v)s_v)^{\alpha_i}]^{1-\theta} - [\rho_d(1 + i_d)]^{\alpha_i(1-\theta)} > \tau_i (F_i^v - F_i^d), \quad (11)$$

where $\tau_i \equiv \frac{\theta^\theta}{(\theta-1)^{\theta-1}} \times \frac{1}{w^{(1-\alpha_i)(1-\theta)}}$.¹³

To analyze the mechanism, we make two simplifying assumptions. First, fixed costs are equal across currencies, $F_i^v = F_i^d$ for all i . Second, the exchange rate s_v is log-normally distributed with mean μ and variance Σ .¹⁴ Under these assumptions, the cutoff condition can be rewritten as:

$$\eta_i^* = 0 \quad \text{iff} \quad \underbrace{\log(1 + i_v) - \log(1 + i_d) + [\mu + 0.5 \alpha_i \Sigma]}_{\text{UIP Deviation}} < \log \left(\frac{\rho_d}{\rho_v} \right), \quad (12)$$

where μ and Σ are the mean and variance of $s_v \sim \text{Log-Normal}(\mu, \Sigma)$. Equation (12) highlights the key insight of the model: (firm-specific) UIP deviations shape importers' currency choices.

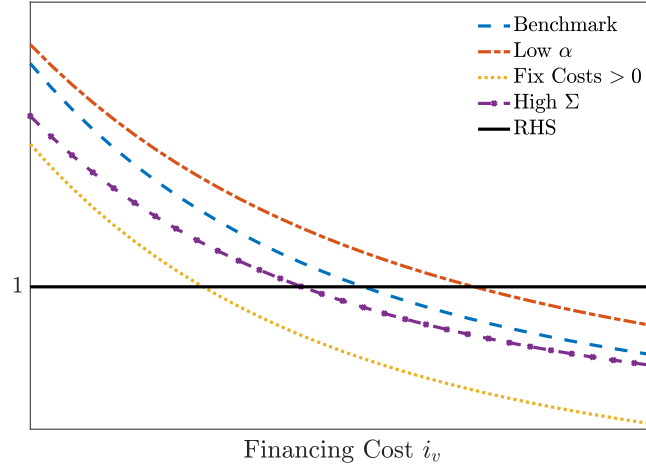
invoicing switching is not driven by specific destination–product pairs within firms. Nonetheless, the model can be extended to multi-market firms, capturing both within- and across-firm externalities.

¹²Without loss of generality, the demand shifter q_d is normalized to one.

¹³See Appendix B for a detailed derivation of Equation (11).

¹⁴While log-normality allows for closed-form solutions, the same intuition holds using a second-order approximation with a general distribution.

Figure 3: Invoicing Choice - Comparative Statics



Note: Figure 3 plots the optimality condition in Equation (11), $\mathbb{E}[(\rho_v(1+i_v)s_v)^{\alpha_i}]^{1-\theta} - \tau_i(F_i^v - F_i^d) > [\rho_d(1+i_d)]^{\alpha_i(1-\theta)}$, for different levels of the vehicle currency interest rate i_v , assuming $s_v \sim \text{Log-Normal}(\mu, \Sigma)$. The RHS, $[\rho_d(1+i_d)]^{\alpha_i(1-\theta)}$, is normalized to one. The blue line shows the benchmark calibration ($w = 0.2$, $\rho_v = \rho_d = 1$, $\theta = 5$, $\mu = 0$, $\Sigma = 1$, $\alpha_i = 0.8$, and $F_i^v - F_i^d = 0$). The other lines illustrate comparative statics relative to the benchmark: lower input share $\alpha_i = 0.7$ (red), higher fixed-cost differential $F_i^v - F_i^d = 1$ (yellow), and higher exchange rate variance $\Sigma = 1.05$ (purple).

This mechanism is novel in the literature and constitutes the main theoretical prediction that we will further validate in the data.

In particular, invoicing imported inputs in vehicle currency v is optimal when its financing costs are sufficiently low to compensate for exchange rate uncertainty (Σ), after accounting for relative input costs (ρ_d, ρ_v). Firms maintain a constant markup over marginal costs, so vehicle-currency invoicing implies that ex-post marginal costs depend on the realized exchange rate, potentially generating costly deviations from the optimal markup. Consequently, firms choose vehicle-currency invoicing only when its financing cost advantage outweighs the risk of markup misalignment. Fluctuations in UIP deviations therefore directly affect firm-level invoicing decisions and, in the aggregate, the dominance of a currency.¹⁵

Next, we analyze how a given interest rate differential across currencies interacts with the firm-specific share of intermediate inputs in production α_i , exchange rate uncertainty Σ , and the differential in currency-specific fixed costs ($F_i^v - F_i^d$).

Figure 3 plots the optimal decision rule in Equation (11) and illustrates how it depends on model parameters. Expected profits under domestic currency financing are normalized to one and independent of i_v (horizontal line). By contrast, expected profits when invoicing working capital in vehicle currency decline with i_v , since higher vehicle interest rates raise marginal costs (downward-sloping line). The intersection of the two curves determines the cut-off value of i_v above which it is no longer optimal to invoice in vehicle currency, that is, $\eta^* = 1$.

¹⁵When $\rho_d = \rho_v$, Equation (12) reduces to the standard UIP condition under log-normality of the exchange rate. In this case, the condition retains the usual no-arbitrage interpretation, though from the perspective of liabilities rather than investments. Traditionally, UIP is a no-arbitrage condition for investors. Here, it governs firms' borrowing costs.

Borrowing Cost Differential and Firm Natural Hedging. Equation (12) shows that the trade-off between cheaper financing option and costly departure from the optimal markup depends on the exposure of the marginal cost to exchange rates, α_i , capturing firm's operational hedging motive (Amiti et al., 2022). *Ceteris paribus*, firms prefer to match their input currency with that of their revenues (e.g. domestic currency), and the incentive is stronger the larger the share of imported inputs (high α_i). Thus, the repayment in foreign currency that makes a firm indifferent between domestic and foreign financing is lower the larger the firm's share of imported inputs. Figure 3 shows that a lower α_i shifts the line representing the expected profits when working capital is invoiced in vehicle currency to the right, making invoicing in vehicle currency more likely. An increase in the variance of the bilateral exchange rate, Σ , raises the likelihood of costly deviations from the optimal constant markup. As a result, the expected profit curve under vehicle-currency invoicing shifts leftward, lowering the probability that firms choose to invoice in vehicle currency.

Borrowing Cost Differential, Fixed Costs, and Strategic Complementarities. The term on the right-hand side of Equation (11) captures how fixed costs of currency use affect the invoicing choice.¹⁶ Assuming that using the vehicle currency has larger fixed costs than the domestic currency ($F^v > F^d$), the firm optimally invoices in vehicle currency only when its financing cost advantage is large enough to offset both the deviation from the optimal constant markup and the higher fixed cost. In line with this reasoning, Figure 3 illustrates that higher fixed cost differential act as a shifter: the expected profit curve for vehicle-currency invoicing moves leftward, making vehicle-currency invoicing less likely.

The dynamics of currency-specific fixed costs are influenced by both strategic complementarities in invoicing and by sunk costs, as shown in Equation (9). A leftward shift in expected profits denominated in vehicle currency can result from either an increase in the fixed cost differential (e.g., first-time adoption due to sunk costs) or from a reduction in the number of competitors using that currency reducing the intensity of strategic complementarities.

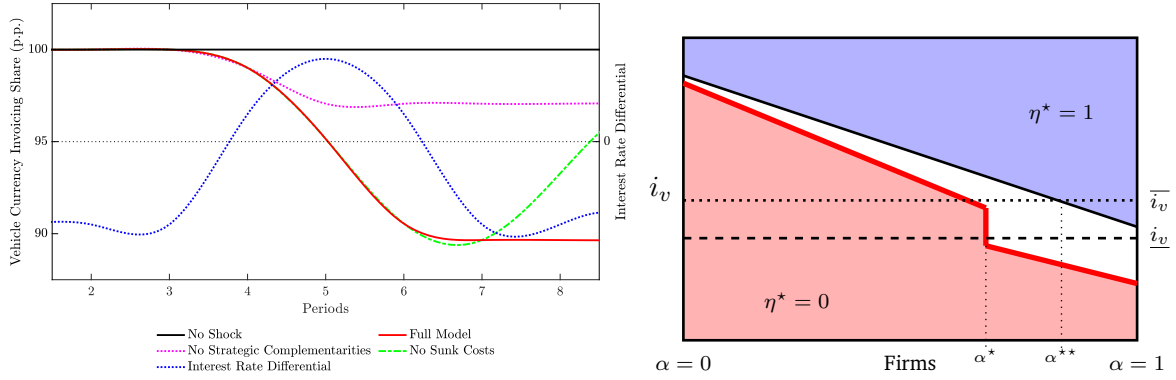
Demand specifications with variable elasticity allows the introduction of strategic complementarities in models where exporters set the invoicing currency of their products (Amiti et al., 2019). In such settings, deviations from the optimal markup in the destination market arise due to fluctuations in the realized export price. Our framework differs from this class of model, as the source of costly fluctuations in the realized markup comes from financing cost on the production side.

4.3 Temporary Shocks, Invoicing Dynamics and Hysteresis

We consider a rise in the cost of finance in vehicle currency i_v – or, equivalently, an increase in UIP deviations – that permanently reverts to its original level after a few periods. We show that our theory can rationalize the main empirical findings of Section 3: the long

¹⁶In deriving the optimal invoicing choice, we assume that firms do not internalize the effect of their choice on the aggregate invoicing share and, consequently, on fixed costs (Equation (9)).

Figure 4: Shock to Vehicle Currency Financing



Note: The right panel plots the optimal invoicing decision for each level of i_v (y-axis) conditional on the level of α (x-axis). The left panel plots the dynamics of the aggregate shares of US dollar invoicing. The red line represents the full model; the green one abstracts away from sunk costs; the purple one abstracts away from strategic complementarities. The blue line represents the dynamics of the relative costs of financing $i_v - i_d$. We use the following symbolic calibration: we normalize ρ_v and ρ_d to one; the elasticity of demand θ is set equal to 5; domestic wage is set to 0.2 (Saravia and Voigtländer, 2012); α_i are uniformly distributed; without loss of generality, we consider the following specification for the fixed costs of invoicing in US dollar: $F^v - F^d = k_0^v - f^d - \gamma^v \omega_{t-1}^v - \gamma_1^v (\omega_{t-2}^v - \omega_{t-1}^v)$, where γ^v is set equal to 0.7, $\gamma^d = 0$, $\gamma_1 = 2.5$, $k_0^v = 1$, and $f^d = \{k_1^d, k_0^d\} = \{0.2, 0\}$; we consider a 3% increase in $i_v - i_d$.

lasting effects on the individual and aggregate import invoicing patterns. Moreover, firms that are larger and more exposed to exchange rate are the first to switch to domestic currency invoicing. Figure 4 displays these results graphically.

The firm-level invoicing patterns presented in Section 3 support the assumption that all firms initially invoice in vehicle currency and have no prior experience invoicing in domestic currency. As a result, firms must pay a sunk cost to switch to domestic currency invoicing. Moreover, we assume that the share of imported goods used in production, α_i , is uniformly distributed: $\alpha_i \sim U[0, 1]$.¹⁷

Short-run response The right panel of Figure 4 graphically represents each firm's invoicing choice. On the horizontal axis, firms are ranked in an ascending order based on their share of imported inputs, α_i . The vertical axis represents the cost of financing in vehicle currency, i_v . For any given level of i_v , there is a threshold $\tilde{\alpha}$ such that firms with $\alpha_i < \tilde{\alpha}$ invoice in domestic currency ($\eta^* = 1$), while firms with $\alpha_i > \tilde{\alpha}$ invoice in vehicle currency ($\eta^* = 0$). The downward-sloped line dividing the box in two areas – domestic (blue area) and vehicle (non-blue area) currency financing – represents the locus of $\tilde{\alpha}$ as a function of i_v .¹⁸

Following the rise in the cost of financing in vehicle currency, larger and more exposed firms are the first to switch to domestic invoicing, consistent with the empirical presented in Section 3. The initial cost of financing in vehicle currency, i_v , is low enough that it is optimal for all firms to invoice in vehicle currency. This is represented by i_v (dash horizontal line) lying below the initial locus of $\tilde{\alpha}$ (solid black line). When i_v increases from the initial level i_v to a higher level $\bar{i}_v$ – meaning foreign invoicing becomes less convenient – a mass of firms

¹⁷The main insights are qualitatively unchanged regardless of the distributional assumption on α_i .

¹⁸For instance, in the special case where $s_v \sim \text{Log-Normal}(\mu, \Sigma)$, the cutoff $\tilde{\alpha}$ is determined by Equation (12) once we solve explicitly for α_i .

with $\alpha_i > \alpha^{**}$ begin borrowing and invoicing in local currency. Notably, the first firms to switch are those with high α_i , which represents a sufficient statistics for both their unhedged exposure to vehicle currency and their market share in the imported inputs market.¹⁹

The presence of strategic complementarities creates a positive feedback in domestic currency invoicing, further expanding its use in imports. The left panel of Figure 4 shows that the aggregate share of imports invoiced in vehicle currency continues to decline even after the cost of financing in vehicle currency, i_v , starts to revert. The use of domestic currency by a group of firms after the shock reduces the fixed cost of invoicing in domestic currency for others, increasing the likelihood that more firms will adopt it. As seen in the data, firms with lower α_i value, relative to the initial switchers, begin to use domestic currency to invoice their imported inputs as domestic invoicing becomes more convenient for them.²⁰

Hysteresis in invoicing The presence of sunk cost in invoicing implies that once a firm switches to domestic currency invoicing, the relative convenience of domestic versus vehicle currency invoicing is permanently altered. As a result, the sunk-cost model generates invoicing hysteresis: even if i_v permanently reverts to the original level $\underline{i}_v$, firms do not switch back to vehicle currency invoicing. Specifically, the difference between the fixed costs of invoicing in vehicle and domestic currency, $F^v - F^d$ in Equation (11), decreases, making domestic currency invoicing more permanently attractive than vehicle currency invoicing.

Graphically, this is represented by a downward shift in the threshold between domestic and vehicle currency invoicing for firms that begin invoicing in domestic currency (i.e. those with $\alpha_i > \alpha^*$). The area where domestic invoicing is optimal ($\eta^* = 1$) extends to include the area above the solid red line (i.e. both the white area and the original blue area). Thus, the new cut-off rule implies that for firms with $\alpha_i > \alpha^*$, it remains optimal to invoice in domestic currency even when ϵ reverts to the original level, $\underline{i}_v$.²¹

At the aggregate level, this implies that the share of imports invoiced in domestic currency does not return to one after i_v reverts to the original level, which is consistent with the observed data. The absence of any partial reversion in the aggregate share of imports invoiced in domestic currency suggests that the sunk cost associated with domestic currency invoicing is significant enough to make the switch permanent. The green dashed line in the left panel shows that, if the sunk cost were smaller, the aggregate share of invoicing could (partially) increase back, as it might become optimal for some firms to revert to vehicle cur-

¹⁹The former follows immediately from the fact that firms are importers only, with revenues denominated in domestic currency. The latter follows from the fact that firms are assumed to have identical size. Thus, higher α_i implies more imported inputs and, thus, a larger market share in the input market.

²⁰In the right panel of Figure 4, firms with $\alpha^* < \alpha_i < \alpha^{**}$ gradually switch to domestic currency invoicing as fixed costs decrease. At the aggregate level, the pink dashed line in the left panel shows that, when the i_v starts to revert, the share of vehicle currency invoicing stops decreasing in the absence of strategic complementarities.

²¹Notice that domestic currency invoicing becomes more appealing even for firms that have not yet paid the sunk cost, i.e. firms with $\alpha_i < \alpha^*$. This is because the fixed cost of invoicing in domestic currency decreases also for them, due to strategic complementarities. Relative to the initial steady state, where all invoicing was in vehicle currency, the fixed cost of invoicing in vehicle currency is now higher since a subset of firms has permanently switched to domestic invoicing. The higher fixed cost of invoicing in vehicle currency F^v makes domestic invoicing more competitive for all firms.

rency invoicing. This would occur if the decrease in the fixed costs of domestic invoicing, F^d , is insufficient to make domestic currency invoicing optimal at the original level of i_v , $\underline{i}_v$.

5 Model Validation

In this section, we test the key predictions of the model against the data. We begin by documenting that the dynamics of the UIP deviation between the Chilean Peso and the US Dollar are consistent with the mechanism that incentivizes firms to switch from US Dollar invoicing to domestic currency invoicing. Next, we show that, as the cost of borrowing in US Dollars rises relative to domestic currency, the probability of invoicing in Chilean Pesos increases with the share of imports invoiced in US Dollars and decreases with the strength of strategic complementarities in US Dollar invoicing. These coefficients are identified using cross-sectional variation across Chilean importers. We then validate the model using an event-study design. Finally, we present evidence consistent with the existence of sunk costs in invoicing.

UIP Deviations Between Chilean Peso and US Dollar. We start by documenting that during the GFC, borrowing in Chilean Peso for a domestic firm was cheaper than borrowing in US Dollars, contrary to the pattern observed in normal times. We construct the UIP deviation between the Chilean Peso and the US Dollar as follows. Define the UIP between the two currencies by:

$$\text{UIP}_{t+1}^{\$, \text{CLP}} = \underbrace{i_t^{\$} - i_t^{\text{CLP}}}_{\Delta \text{Interest Rates}} + \underbrace{s_{t+1}^e - s_t}_{\Delta \text{Exchange Rate}}, \quad (13)$$

where $i_t^{\$} \approx \log(1 + i_t^{\$})$ and $i_t^{\text{CLP}} \approx \log(1 + i_t^{\text{CLP}})$ denote commercial lending rates for short-term (less than one year) loans in US Dollars and Chilean Pesos, respectively. Here, s_t is the log spot exchange rate (CLP per USD) and s_{t+1}^e is the log expected exchange rate one year ahead. A negative UIP deviation ($\text{UIP}_{t+1} < 0$) indicates that borrowing in Chilean Pesos is more costly than borrowing in US Dollars for a domestic firm, while a positive value implies the opposite. ²²

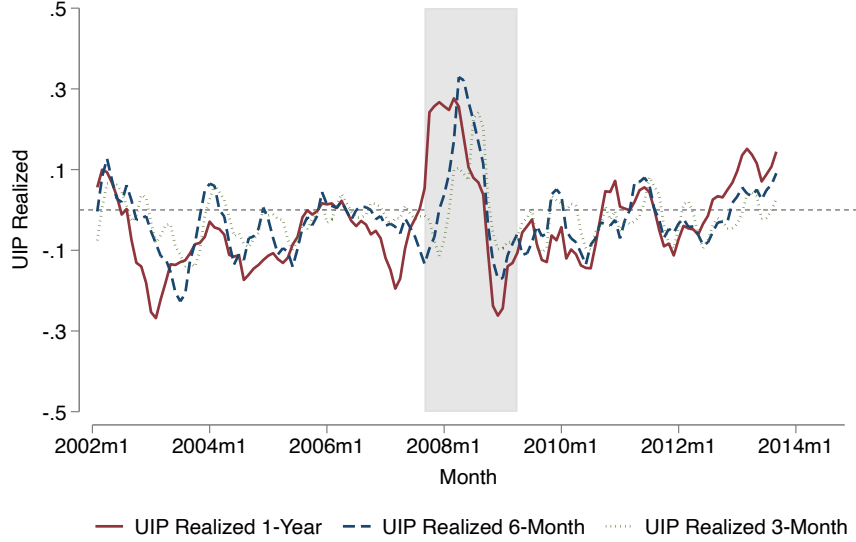
Figure 5 shows that during the GFC, the cost of financing in US Dollars became relatively more expensive than in domestic currency. The data on bank lending rates come from the Chilean financial authority responsible for supervising the banking system. The red line plots the dynamics of the realized UIP deviation defined in Equation (13), where the expected bilateral exchange rate is replaced by the realized one-year-ahead exchange rate between the Chilean Peso and the US Dollar. The blue dashed line and the gray dotted line report the same measure using the six-month and three-month realized exchange rates, respectively.

In the full sample, both the mean and the median of the realized UIP are negative, equal

²²Equation (13) follows from the standard no-arbitrage condition. If a firm borrows one Chilean Peso today, it repays $1 + i_t^{\text{CLP}}$ Chilean Peso next period. If it borrows one US Dollar, the firm obtains $1/s_t$ Chilean Pesos today, which requires repayment of $\frac{s_{t+1}^e}{s_t}(1 + i_t^{\$})$ US Dollar in the next period. Borrowing in CLP is more expensive if $1 + i_t^{\text{CLP}} > \frac{s_{t+1}^e}{s_t}(1 + i_t^{\$})$. Taking logs and rearranging yields Equation (13).

to -0.04 and -0.03 , respectively, indicating that borrowing in Chilean Pesos is typically more expensive than borrowing in US Dollars. However, during the GFC the average realized UIP turns positive, equal to 0.07 , implying that US Dollar borrowing became more costly in this period. This transitory reversal is consistent with the key theoretical mechanism needed to explain the observed decline in vehicle-currency invoicing.²³

Figure 5: Realized UIP Between the US Dollar and Chilean Peso



Note: Figure 5 plots the one-year realized UIP deviation between the Chilean Peso and the US Dollar, constructed from Equation (13). A positive value indicates that borrowing in Chilean Pesos is relatively cheaper than borrowing in US Dollars. Interest rates are measured using commercial lending rates on short-term (less than one year) loans denominated in US Dollars and Chilean Pesos. Expected bilateral exchange rates are replaced with the corresponding one-year-ahead realized exchange rates.

Strategic Complementarity in Invoicing and Operational Hedging. Here we define the construction of the two main determinants of invoicing highlighted in the model: strategic complementarities in invoicing and the share of imports denominated in US Dollars, which we interpret as operational or natural hedging.²⁴ First, we measure strategic complementarities in invoicing as in Crowley et al. (2020):

$$SC_{(-f)ost}^{\$} = \frac{\sum_{k \neq f} \text{ToT Imp}_{kost}^{\$}}{\sum_c \sum_{k \neq f} \text{ToT Imp}_{kost}^c} \quad (14)$$

²³The dynamics of borrowing costs invoicing in different currencies in Chile are in line with previous evidence from Vial et al. (2020) and Betancour et al. (2006). Moreover, changes in bank or capital market regulations do not align with the timing or the effects of the phenomena we document. The 2010 banking reform aimed at liberalizing the financial sector, increasing transparency and supervision, and deepen global integration - e.g. providing better access to FX hedging instruments (Fund, 2010). Similarly, Chile updated its prudential framework after other advanced economies and OECD countries, Madeira and Olivares (2021) and <https://www.sipa.columbia.edu/sites/default/files/2023-02/Macroprudential-Policies-A-View-From-Chile.pdf>. Basel III was in fact implemented in 2019 (<https://www.cmfchile.cl/portal/principal/613/w3-article-50324.html>).

²⁴Under CRS Cobb–Douglas technology, the share of total costs spent on intermediate inputs is constant at α_i . However, because we do not observe firm-level cost data, we cannot directly measure total variable costs. Instead, as is standard in the literature (e.g., Crowley et al. (2020)), we approximate this by the share of total inputs invoiced in US Dollars.

where $(-f)$ is the set of competitors of firm f in sector s (HS6) who import from origin o at time t . Equation (5) measures the share of total imports of competitors at the industry (HS6) and origin level invoiced in US Dollar. Clearly, the invoicing choice of firm f is endogenous to the choice of its competitors $(-f)$. Thus, we construct the following instrumental variables to correct for the existence of possible endogeneity.

$$\widehat{SC}_{(-f)ost}^{\$} = \sum_{k \neq f} \frac{S_{kost}}{1 - S_{kost}} \times \underbrace{\frac{\text{ToT Exp}_{kt}^{\$}}{\sum_c \text{ToT Exp}_{kt}^c}}_{NH_{kt}^{\$}}, \quad (15)$$

where $S_{kost} = \frac{\text{ToT Imp}_{kost}}{\sum_f \text{ToT Imp}_{fost}}$ is therefore the import market share of firm k from origin o , industry (HS6), and time t level among all Chilean importers. The term $NH_{kt}^{\$}$ corresponds to the measure of operational hedging, defined as the share of firm k 's total exports that are invoiced in US Dollars.

5.1 Firm-Level Evidence

Using both an event study approach and cross-sectional regression analysis, we test whether the differential between US dollar and domestic financing costs is a determinant of invoicing choice. Finally, we provide evidence supporting the presence of sunk costs in invoicing decisions, which explains the observed hysteresis in the data.

Cross-Sectional Analysis. We start by estimating the following specification:

$$\begin{aligned} \text{Pr(PESO)}_{fpot} = & \beta_0 + \beta_1 \text{UIP}_t^{\$,CLP} + \beta_2 NH_{kt}^{\$} + \beta_3 \widehat{SC}_{(-f)ost}^{\$} + \\ & + \beta_4 NH_{kt}^{\$} \times \text{UIP}_t^{\$,CLP} + \beta_5 \widehat{SC}_{(-f)ost}^{\$} \times \text{UIP}_t^{\$,CLP} + \\ & + FE + \nu_{foit}, \end{aligned} \quad (16)$$

where the dependent variable is a dummy variable that takes value equal one when the transaction at the f_{opt} level is invoiced in Chilean Peso and zero if in US Dollar.

Table 3: Model Validation: Cross-Sectional Analysis

	OLS		IV	
	(1) Pr(PESO) _{f opt}	(2) Pr(PESO) _{f opt}	(3) Pr(PESO) _{f opt}	(4) Pr(PESO) _{f opt}
UIP _{t+1} ^{CLP}	0.028*** (0.001)	0.026*** (0.001)	–	–
SC _{(-f)ost} ^{\$}		-0.033*** (0.002)	-0.192*** (0.005)	-0.121*** (0.003)
NH _{ft} ^{\$}		-0.017*** (0.001)	-0.015*** (0.001)	-0.015*** (0.001)
SC _{(-f)ost} ^{\$} × UIP _{t+1} ^{CLP}			-0.170*** (0.012)	-0.165*** (0.011)
NH _{ft} ^{\$} × UIP _{t+1} ^{CLP}			-0.045*** (0.014)	-0.054*** (0.015)
Firm-Origin-HS8	YES	YES	YES	NO
Year	NO	NO	YES	YES
Firm-Origin	NO	NO	NO	YES
Firm-HS8	NO	NO	NO	YES
Observations	1001124	965928	965928	1158217

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3 reports the estimates from Equation (16). In Columns (1) and (2), the coefficient on the UIP differential, β_1 , is positive and significant, with magnitudes of 0.028 and 0.026, respectively. This implies that a one–percentage point increase in the UIP gap, defined in equation (13), raises the probability of invoicing in Chilean Peso by about 2.6–2.8 percentage points. This result aligns with the model’s prediction that a rise in the expected cost of borrowing in US Dollar relative to the Chilean Peso increases the incentive to invoice in the domestic currency.

Columns (3) and (4) include interaction terms capturing operational hedging ($\text{NH}_{ft}^{\$}$) and strategic complementarities in invoicing ($\text{SC}_{(-f)ost}^{\$}$). Both interactions with $\text{UIP}_{t+1}^{\$,CLP}$ are negative and precisely estimated. Consistent with the main predictions of the model, this implies that, conditional on the US Dollar becoming more expensive, firms are less likely to switch to peso invoicing the more their competitors invoice their imports in US Dollars and the more they invoice their own exports in US Dollars. The first result suggests that strategic complementarities in invoicing reduce the incentive to switch, while the second indicates that firms with greater export invoicing in US Dollars seek to match their import invoicing currency, thus lowering their incentive to switch.

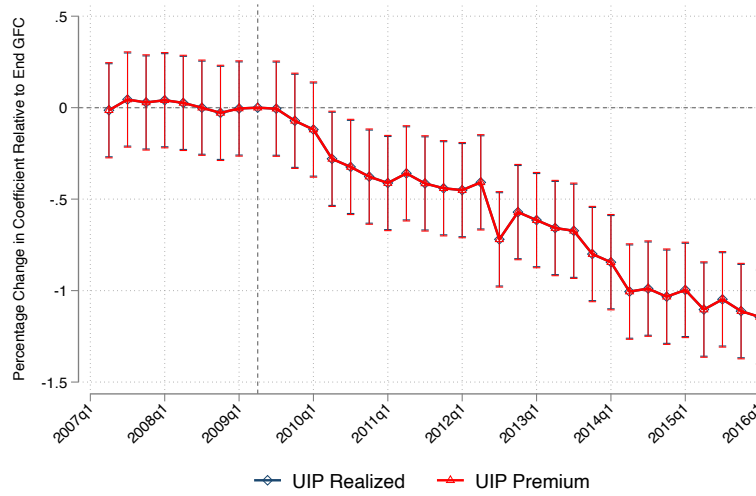
Event Study Design. We estimate the following specification:

$$\text{Pr(PESO)}_{f opt} = \sum_{s=2007Q1}^{2016Q4} \left[\mathbf{1}[t = s] \times \left(\beta_s \cdot \text{SC}_{(-f)ost,2007Q1}^{\$} \cdot \text{UIP}_{GFC}^{\$,CLP} \right) \right] + \gamma_{fpo} + \epsilon_{fpot} \quad (17)$$

where $SC_{(-f)os,2007Q1}^{\$}$ is the sector–origin measure of strategic complementarity in USD invoicing for firm f (HS6) measured in 2007Q1, and $UIP_{GFC}^{\$,CLP}$ is the average UIP deviation between USD and CLP over the NBER recession window (2007Q4–2009Q2). The indicator $1[t = s]$ picks out each quarter s and γ_{fpo} denotes firm $\times$ product $\times$ origin fixed effects. Hence β_s gives the marginal effect (in probability points) at quarter s of a one-unit increase in the pre-GFC strategic-complementarity $\times$ UIP interaction on the probability of invoicing in USD.

The theory predicts $\beta_s > 0$ when UIP is positive (borrowing in USD is more expensive), because strategic complementarity makes firms less likely to switch away from USD invoicing. Over time, as fewer firms invoice in USD, the importance of strategic complementarity should decline, so β_s is expected to fall after the GFC.

Figure 6: Event Study - Strategic Complementarity in Invoicing



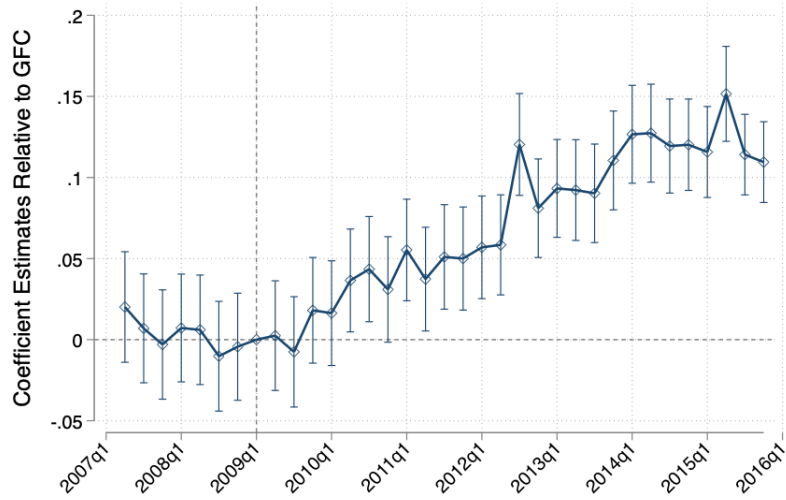
Note: The Figure plots the coefficients β_τ from the event study specification in Equation (??), capturing the differential increase in Chilean Peso invoicing by more exposed firms. Coefficient in 2007 is normalized to zero. Exposure is measured by: share of imports denominated in US dollars over the total imports of the firm. Data are from [Garcia-Marin et al. \(2019\)](#). Standard errors are clustered at firm level.

We also modify the previous specification in the following way

$$\Pr(\text{PESO})_{f\text{opt}} = \sum_{s=2007Q1}^{2016Q4} \left[1[t = s] \times \left(\beta_s \cdot \alpha_{f,2007Q1} \cdot UIP_{GFC}^{\$,CLP} \right) \right] + \gamma_{fpo} + \epsilon_{f\text{pot}} \quad (18)$$

where $\alpha_{f,2007Q1}$ is the share of total imports denominated in US Dollar for firm f in the first quarter of 2007. Similar to before, β_s gives the marginal effect (in probability points) at quarter s of a one-unit increase in the pre-GFC firm-USD Exposure $\times$ UIP interaction on the probability of invoicing in USD. The theory predicts that, conditional on the cost of borrowing in US Dollars being higher, firms with greater import exposure to the US Dollar before the GFC are more likely to invoice in Chilean Pesos. Figure 7 confirms the prediction of the model.

Figure 7: Event Study - USD Exposure



Note: The Figure plots the coefficients β_τ from the event study specification in Equation (??), capturing the differential increase in Chilean Peso invoicing by more exposed firms. Coefficient in 2007 is normalized to zero. Exposure is measured by: share of imports denominated in US dollars over the total imports of the firm. Data are from [Garcia-Marin et al. \(2019\)](#). Standard errors are clustered at firm level.

On Sunk Costs. A key element of the theoretical framework to explain the hysteresis in invoicing choices is the presence of sunk cost in invoicing. We follow the international trade literature in providing evidence consistent with the presence of sunk costs in invoicing. The key implication of sunk costs is that the decision of invoicing in a specific currency increases when the same currency has been previously used. Therefore, conditional on the other forces determining invoicing choices, the past invoicing share should be a predictor of current invoicing choices. This idea translates in the following econometric specification:

$$y_{f,oit}^{\$} = \beta y_{f,t-1}^{\$} + Controls_{f,oit} + FE + \nu_{f,oit}, \quad (19)$$

where $y_{f,oit}^{\$}$ is the (log) share of US dollar invoicing or a dummy variable equal to 1 when US dollar invoicing is used and 0 otherwise, $y_{f,t-1}^{\$}$ is the lagged (log) share of US dollar invoicing at the firm level, and β is the coefficient of interest.

The presence of the lagged endogenous variable on the right side poses potential endogeneity concerns due to highly serially correlated unobserved characteristics that induce persistence in invoicing choices, potentially bias upward the coefficient of interest β . We follow the literature and, in addition to standard controls such natural hedging and strategic complementarities ([Amiti et al., 2022](#)), we saturate the specification with fixed effects ([Bernard and Jensen, 2004](#); [Timoshenko, 2015](#); [Das et al., 2007](#)).²⁵ We alternatively estimate Equation

²⁵We introduce a natural hedging measure defined as the share of export invoiced in US dollars ([Crowley et al., 2020](#)), and a strategic complementarities measure as the average share of imports invoiced in US dollars among all competitors in a given sector(HS6)-origin pair [Amiti et al. \(2022\)](#). We also construct an instrument for the latter following [Crowley et al. \(2020\)](#): we use the import-weighted average US dollar exposure of all competitors in a given sector(HS6)-origin pair, where the exposure is defined as the difference between the total exports denominated in US dollars and the total imports denominated in US dollars normalized by the firm's total trade.

(19) in first difference via OLS and via an Arellano-Bond GMM estimator using lagged levels of the right-side variables as instruments.

Table 4: Testing the Model - Sunk Cost

	Dummy USD					Share USD	
	(1) OLS	(2) OLS	(3) OLS	(4) IV	(5) IV	(6) First Difference - IV	(7) First Difference - GMM
Lagged Dollar Share	0.216 (0.009)	0.126 (0.006)	0.040 (0.005)	0.040 (0.005)	0.157 (0.046)		
Lagged Δ Dollar Share						0.036 (0.017)	-0.056 (0.011)
Controls	No	No	Yes	Yes	Yes	Yes	Yes
Origin $\times$ Year & Product $\times$ Year	No	Yes	Yes	Yes	No	No	No
Firm $\times$ Product $\times$ Origin & Year	No	No	No	No	Yes	Yes	No
Observations	2187942	2181688	774868	774868	623768	356394	260203

Note: The Table reports the coefficients from specification in Equation (19). The dependent variable in specifications (1) to (5) is a dummy variable equal to 1 when US dollar invoicing is used and 0 otherwise; The dependent variable in specifications (5) to (7) is the (log) share of US dollar invoicing. Column (6) is estimated in first difference via OLS. Column (7) is estimated using an Arellano-Bond GMM estimator using lagged levels of the right-side variables as instruments. "Lagged Dollar Share" refers to the lagged (log) share of US dollar invoicing at the firm level. Controls include: a natural hedging measure defined as the share of export invoiced in US dollar; a strategic complementarities measure as the average share of imports invoiced in US dollar among all competitors in a given sector(HS6)-origin pair. IV columns refers to the case in which the strategic complementarity measure is instrumented with the import-weighted average US Dollar exposure of all competitors in a given sector(HS6)-origin pair. Standard errors are clustered at firm level. We use data from official customs 2007-2019.

Table 4 provides strong support for the presence of sunk costs in invoicing. The benchmark specifications in Column (4) and (5) show that a 1% increase in the past share of US dollar invoicing increases the probability of invoicing by 4% and the current share of US dollar invoicing by 0.16%, respectively. Alternative specifications using first difference also show the relevance of past US dollar invoicing share for the contemporaneous invoicing decision. In Appendix A, Table 13 shows that the results are robust to the use of lagged (log) share of US dollar invoicing at the firm-origin-product level, rather than firm level, suggesting that these costs exhibit also some product-origin specific component.

6 Aggregate Implications

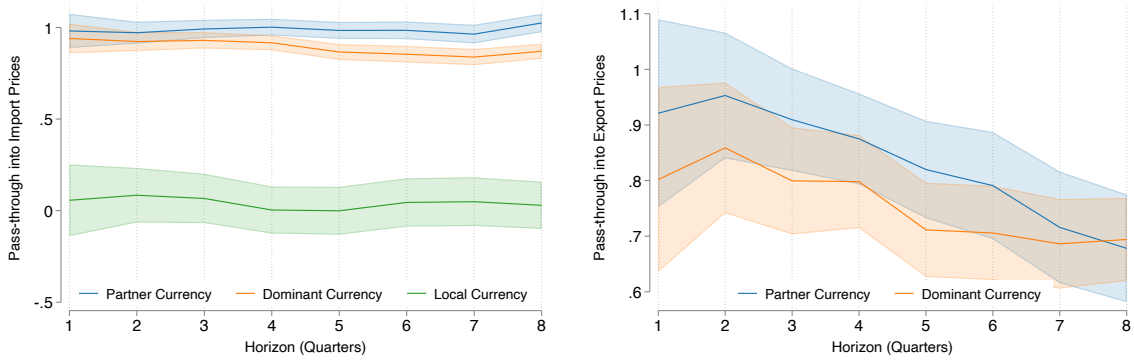
In this section we quantitatively assess the macroeconomic implications of the large changes in invoicing shares after the GFC. We focus on the implications of invoicing choices for the dynamics of terms of trade and trade balance. We proceed as follow. We first estimate short-run and long-run price and quantities elasticities to exchange rate fluctuations accounting for the different invoicing currencies, building on existing empirical frameworks (Adler et al., 2020; Barbiero, 2021; Chen et al., 2022). We then leverage the estimated elasticities and the evolution of invoicing shares to conduct accounting exercises (Auer et al., 2021). We find that the change in the invoicing shares after the GFC implies a trade balance elasticity to exchange rate which is 30% higher.

Price and Quantity Elasticities to Exchange Rate We first estimate the exchange rate pass-through to border prices and quantities, both in the short-run and in the long-run, conditioning on the currency of invoicing. Following existing econometric frameworks such as [Adler et al. \(2020\)](#), the benchmark specification is:

$$\begin{aligned} \Delta P_{f,opt}^k = & \sum_{l=0}^8 \beta_l^{y,CLP} \Delta e_{t-l}^{CLP/o} \times D_j^{LCP} + \sum_{l=0}^8 \beta_l^{y,p} \Delta e_{t-l}^{CLP/o} \times D_j^{PCP} + \\ & + \sum_{l=0}^8 \beta_l^{y,D} \Delta e_{t-l}^{CLP/\$} \times D_j^{DCP} + \sum_{l=0}^8 \beta_l^{y,D} \Delta e_{t-l}^{\$/o} \times D_j^{DCP} + \\ & + \alpha_j + \phi x_{jt} + \delta_{t \times \Delta} + \varepsilon_{jt}, \end{aligned} \quad (20)$$

where $\Delta P_{f,opt}^k$ denotes the log change in unit values (expressed in Chilean pesos) of product p imported by firm f from origin o between quarters $t - 1$ and t , invoiced in currency k . An increase in $\Delta e_t^{CLP/o}$ corresponds to a depreciation of the Chilean peso relative to the currency of origin country o . The dummy variables D_j^{LCP} , D_j^{PCP} , and D_j^{DCP} indicate whether the transaction is invoiced in local, producer, or dominant currency, respectively. Because of the triangular relationship among exchange rates, for dominant currency-priced products we decompose the bilateral peso-partner exchange rate, $\Delta e^{CLP/o}$, into two components capturing sensitivity to peso-dollar fluctuations and partner-dollar fluctuations ([Barbiero, 2021](#); [Chen et al., 2022](#)). For a given invoicing scheme, short-run ERPT coefficients are equal to β_0 while cumulative ERPT at horizon s are equal to $\beta_s = \sum_{l=0}^s \beta_l$.

Figure 8: Price Sensivities to Exchange Rates



Note: The left (right) panel show the estimated coefficients from the specification in Equation (20) using import (export) prices as dependent variable. We use data from official customs running from 2007 to 2019 for imports and 2009 to 2019 for exports. We exclude the LCP case for export prices because of the absence of sufficient observations in the estimation.

Figure 8 plots the estimates of the cumulative exchange rate pass-through into import (left panel) and export (right panel) prices obtained from the main specification in Equation (20). We show that exchange rate pass-through varies substantially across invoicing choices, both in the short-run and in the long-run. Pass-through into import prices is zero when transactions are invoiced in local currency (Chilean peso), while it is complete when transactions are invoiced in dominant and partner currencies. Pass-through rates into import prices remain

at essentially the same level after two years. On the export side, pass-through rates are almost complete in the short-run when transactions are invoiced in dominant and partner currencies (80% and 90%, respectively).²⁶ In the long-run, the pass-through rate decreases to approximately 70% in both cases. These patterns are in line with previous findings and with the idea that prices are sticky in the currency in which they are invoiced (Chen et al., 2022; Gopinath et al., 2010).

Figure 20 in Appendix A reports the pass-through rates into import and export quantities. We show that different pass-through rates into prices translates into different response of quantities, providing evidence of allocative effects of invoicing choices (Amiti et al., 2022). Import quantities do not react in the short-run, independently of the invoicing currency. They decrease by 20% in the long-run when invoiced in dominant or partner currencies.²⁷ The response of quantities remains essentially unchanged in the long-run when invoiced in local currency, consistent with the fact that, after a depreciation, prices invoiced in Chilean pesos do not adjust. The sluggish response of quantities is in line with expenditure switching forces toward domestic goods.

Thus, abstracting from invoicing choices improperly informs on the effects of exchange rate fluctuations on the macroeconomy (Gopinath and Itskhoki, 2022; Barbiero, 2021). We now explore the quantitative relevance focusing on the dynamics of the terms of trade.

On the Effects on Trade Balance In traditional models of international economics (e.g. Mundell-Fleming), in which international prices are invoiced in the exporter's currency, exchange rates play a key role in external adjustment. Exchange rate changes induce movements in imported and exported quantities by influencing the relative price of domestic and foreign goods. However, the widespread use of US dollar invoicing, together with the large differences in price and quantity elasticities documented above, impacts how the trade balance responds to exchange rate fluctuations. We focus on the dynamics of a country's external imbalance following a movement of its exchange rate vis-a-vis all other currencies, and how invoicing patterns at the aggregate level influence its response to exchange fluctuations.

Let define the trade balance of Chile as the same of net export from the rest of the world, as follow: $TB_t = \sum_{i \in I} (P_{it}^X Q_{it}^X - P_{it}^M Q_{it}^M)$, where P_{it}^X and Q_{it}^X (P_{it}^M and Q_{it}^M) are the price and quantity of exports to (import from) country i , respectively. It follows that the effect of exchange rates on the trade balance (relative to GDP, Y_t) at horizon l is given by:

$$\frac{\Delta TB_t^l}{Y_t} = \Delta e \times \frac{X_t}{Y_t} \left[\sum_{\tau=0}^l (\beta_{\tau}^{PX,CLP} + \beta_{\tau}^{QX,CLP}) S^{X,CLP} + (\beta_{\tau}^{PX,D} + \beta_{\tau}^{QX,D}) S^{X,D} + (\beta_{\tau}^{PX,p} + \beta_{\tau}^{QX,p}) S^{X,p} \right] \quad (21)$$

²⁶We do not report the pass-through rate into export prices conditional on being invoiced in Chilean peso because of very noisy estimates due to a very restricted sample of only approximately 500 observations.

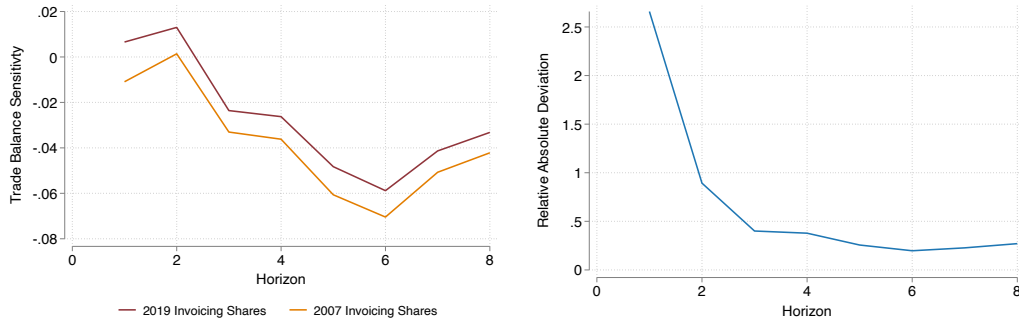
²⁷These magnitudes are smaller compared to the estimates from the macro (Boehm et al., 2023) or trade literature (Broda and Weinstein, 2006). Nevertheless, the quantity response may be muted by the fact that we are working with customs prices, without considering additional rounds of incomplete pass-through into final consumer prices (Auer et al., 2021).

$$- \Delta e \frac{M_t}{Y_t} \left[\sum_{\tau=0}^l (\beta_{\tau}^{PM,CLP} + \beta_{\tau}^{QM,CLP}) S^{M,CLP} + (\beta_{\tau}^{PM,D} + \beta_{\tau}^{QM,D}) S^{M,D} + (\beta_{\tau}^{PM,p} + \beta_{\tau}^{QM,p}) S^{M,p} \right], \quad (22)$$

where $S^{X,z}$ ($S^{M,z}$) the share of exports (imports) invoiced in currency z , and $\beta_{\tau}^{j,z}$ is the elasticity of j to exchange rates when invoiced in currency z at horizon τ , with j being import and export prices and quantities ($\{PX, PM, QX, QM\}$) and $z \in \{CLP, p, D\}$. In other words, the aggregate response of trade balance to a movement of the exchange rate vis-a-vis all other currencies is a weighted sum of the response of import and export prices and quantities, weighted by their respective invoicing shares.

Figure 9 shows that the large increase in the share of Peso invoicing after the GFC lowers the sensitivity of the Chilean trade balance at all horizons. In 2019, trade balance is approximately 30% less sensitive in the long-run, while the sensitivity flips sign and become positive in the short-run. The key driver is the fact that imports denominated in Chilean peso exhibit a lower sensitivity at all horizons, especially in the short run. The lower sensitivity of trade balance raises questions about the benefits of exchange rate flexibility.²⁸

Figure 9: Trade Balance Sensitivity to Exchange Rates



Notes: The left panel plots the 2-periods moving average of the response of the trade balance to a depreciation of the Chilean peso computed from the formulas in Equation (22). We use the estimated coefficients from Equation (20). Invoicing shares are computed from official import (export) customs data from 2007 (2009) for the initial period (orange line) and from 2019 for the last period (red line). The share of export and import on GDP is computed from IMF data. The right panel plots the difference between the response of the trade balance in 2007 and in 2019 relative to the 2019 case.

On the Effects on Terms of Trade and Import Inflation We follow the same steps to assess how the sensitivity of import inflation and the terms of trade evolved after the GFC. Figure 22 in Appendix A shows that the import price index of Chile becomes less sensitive to exchange rate fluctuations by approximately 5% at all horizon, as a larger share of imports is non-sensitive to exchange rate fluctuations. Similarly, Figure 21 in Appendix A shows how lower import price sensitivity translates into a lower sensitivity for the terms of trade. In this case, following a depreciation, the terms of trade decrease by less in 2019 compared to 2007, with a larger discrepancy in the short-run.

²⁸Notice that the trade balance does not follow the standard J-curve dynamic after a depreciation of the exchange rate. This is a consequence of the Dominant Currency Paradigm, which departs from the producer-pricing assumption of the traditional models of international economics.

Conclusion

Our analysis reveals how temporary shocks to dollar financing availability can weaken the dominance of the US dollar in international trade, challenging the historical notion that abrupt shifts in the international monetary system require permanent changes in economic conditions. We document a gradual decline in the share of Chilean imports invoiced in US dollars over a decade, accompanied by a corresponding increase in Chilean peso invoicing. Decomposition exercises show that surviving firms contribute approximately 80% of the observed shift, with larger and more exposed firms leading the change.

Our model explains firms' invoicing decisions based on the costs of financing in domestic versus vehicle currencies, treating imported inputs as working capital. We show that temporary shocks to financing costs or UIP deviations influence invoicing patterns, with sunk costs introducing hysteresis, permanently altering the convenience of using different currencies.

We test the key mechanisms of our theory using firm-level data and various econometric techniques. Our findings support the key mechanisms and the role of strategic complementarities and sunk costs, explaining the observed invoicing patterns in Chilean imports.

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A Appendix: Data & Robustness

A.1 Transition Matrices

Table 5: Yearly Transition Matrices Before and After the GFC at the *FPOT* Level

Before GFC				
From → To	Other	USD	EUR	CLP
Other	84.27	10.65	4.92	0.17
USD	0.42	98.48	1.06	0.04
EUR	0.34	4.18	95.46	0.02
CLP	1.03	7.06	1.72	90.19
After GFC				
From → To	Other	USD	EUR	CLP
Other	84.93	12.04	2.76	0.27
USD	0.33	98.55	0.94	0.18
EUR	0.40	4.27	94.80	0.52
CLP	0.38	8.82	1.81	88.99

Notes: Table 2 reports the results from the dynamic Olley–Pakes decomposition with entry and exit at the origin–product level for Chilean imports between 2007 and 2019. It corresponds to the decompositions in Equations (1), (2), and (3). Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year.

Table 6: Yearly Transition Matrices Before and After the GFC at the *FOT* Level

Before GFC				
From → To	Other	USD	EUR	CLP
Other	74.51	18.23	7.24	0.02
USD	0.67	97.71	1.59	0.03
EUR	0.62	6.49	92.88	0.01
CLP	4.35	15.94	2.90	76.81
After GFC				
From → To	Other	USD	EUR	CLP
Other	76.21	19.08	4.52	0.19
USD	0.59	97.79	1.49	0.13
EUR	0.55	6.29	93.04	0.13
CLP	0.75	19.30	4.64	75.31

Notes: Table 6 reports yearly transition probabilities across invoicing currencies at the FOT level, before (2004–2008) and after (2009–2017) the Global Financial Crisis (GFC). Values are expressed as percentages and rows sum to 100.

Table 7: Yearly Transition Matrices Before and After the GFC at the *FT* Level

Before GFC				
From → To	Other	USD	EUR	CLP
Other	58.03	34.58	7.39	0.00
USD	0.92	97.09	1.93	0.06
EUR	1.58	19.31	79.08	0.03
CLP	0.00	47.62	4.76	47.62
After GFC				
From → To	Other	USD	EUR	CLP
Other	59.85	34.13	5.97	0.04
USD	0.78	96.40	2.70	0.13
EUR	0.98	17.53	81.40	0.10
CLP	0.72	46.04	8.63	44.60

Notes: Table 7 reports yearly transition probabilities across invoicing currencies at the Firm–Year level, before (2004–2008) and after (2009–2017) the Global Financial Crisis (GFC). Values are expressed as percentages and rows sum to 100.

Table 8: Yearly Transition Matrices Before and After the GFC at the *FOPT* Level (Top Quartile by Transaction Value)

Before GFC				
From → To	Other	USD	EUR	CLP
Other	88.41	7.34	4.03	0.21
USD	0.24	99.05	0.68	0.03
EUR	0.20	2.89	96.90	0.01
CLP	0.86	4.31	0.00	94.83
After GFC				
From → To	Other	USD	EUR	CLP
Other	89.43	8.75	1.63	0.20
USD	0.17	99.16	0.52	0.14
EUR	0.19	2.78	96.72	0.31
CLP	0.23	6.68	1.47	91.63

Notes: Table 8 reports yearly transition probabilities across invoicing currencies at the FOPT level, conditional on the FOB transaction value being in the top quartile ($\geq p75$). Results are shown before (2004–2008) and after (2009–2017) the Global Financial Crisis (GFC).

A.2 Additional Decompositions

Table 9: Dynamic Olley-Pakes Decomposition with Entry and Exit: Imports at Origin-Product Level

	USD	Peso	Euro	Others
-				
Aggregate Invoicing Share	-9.452	6.093	3.611	-0.251
Contribution Net Entry				
Total	-1.769	0.778	0.813	0.178
Firm Entry	-2.618	0.856	1.872	-0.110
Firm Exit	0.850	-0.078	-1.059	0.288
Contribution Incumbents				
Total	-7.683	5.315	2.798	-0.430
Within Firm	-4.560	6.367	-1.337	-0.470
Between Firm	-3.123	-1.052	4.134	0.041
Within Firm				
Total	-4.560	6.367	-1.337	-0.470
Net Entry Origin	-0.489	0.875	-0.188	-0.198
Within Origin	-4.801	5.192	-0.213	-0.177
Between Origin	0.729	0.301	-0.935	-0.095

Notes: Table 9 reports the results from the dynamic Olley-Pakes decomposition with entry and exit at the origin-industry level for Chilean imports between 2007 and 2020. It corresponds to the decompositions in Equation (1), (2), and (3). Industry are defined at the HS2 level. One time period corresponds to one year.

Table 10: Dynamic Olley-Pakes Decomposition with Entry and Exit: Imports at Origin Level

	USD	Peso	Euro	Others
-				
Aggregate Invoicing Share	-9.452	6.093	3.611	-0.251
Contribution Net Entry				
Total	-1.769	0.778	0.813	0.178
Firm Entry	-2.618	0.856	1.872	-0.110
Firm Exit	0.850	-0.078	-1.059	0.288
Contribution Incumbents				
Total	-7.683	5.315	2.798	-0.430
Within Firm	-4.560	6.367	-1.337	-0.470
Between Firm	-3.123	-1.052	4.134	0.041
Within Firm				
Total	-4.560	6.367	-1.337	-0.470
Net Entry Origin	0.634	0.080	-0.222	-0.491
Within Origin	-6.118	5.589	0.435	0.095
Between Origin	0.924	0.699	-1.549	-0.074

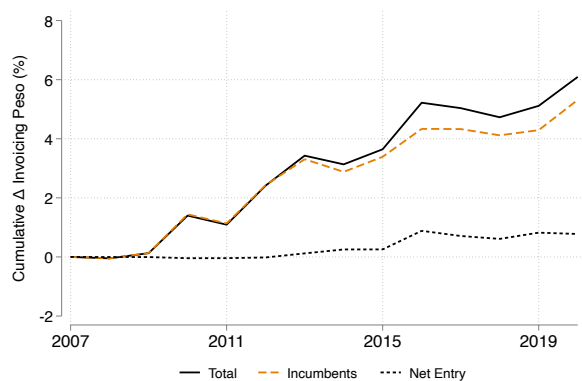
Notes: Table 10 reports the results from the dynamic Olley-Pakes decomposition with entry and exit at the origin level for Chilean imports between 2007 and 2020. It corresponds to the decompositions in Equation (1), (2), and (3) where Equation (3) is properly adjust to consider only the origin level. One time period corresponds to one year.

Table 11: Olley-Pakes Decomposition of Exports Invoicing

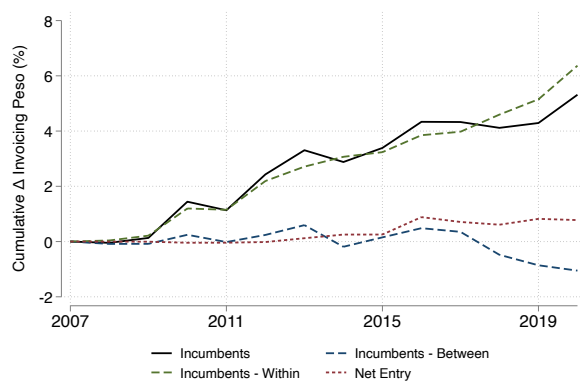
	USD	Peso	Euro	Others
Aggregate Invoicing Share	0.488	0.009	-0.949	0.452
Net Entry	0.015	-0.000	-0.230	0.215
Firm Entry	-0.305	0.012	0.087	0.205
Firm Exit	0.319	-0.013	-0.317	0.010
Contribution Incumbents	0.473	0.009	-0.719	0.237
Within Firm	0.717	-0.326	-0.472	0.081
Between Firm	-0.244	0.335	-0.247	0.156
Within Firm				
Net Entry Destination	-1.551	2.107	-0.528	-0.028
Within Destination	0.932	-2.445	1.272	0.242
Between Destination	1.336	0.013	-1.215	-0.133

Notes: The Table reports the results from the dynamic Olley–Pakes decomposition with entry and exit at the origin–product level for Chilean exports between 2007 and 2019. It corresponds to the decompositions in Equations (1), (2), and (3). Products are defined at the HS8 level, the most granular classification available in the dataset. We report the cumulative decomposition.

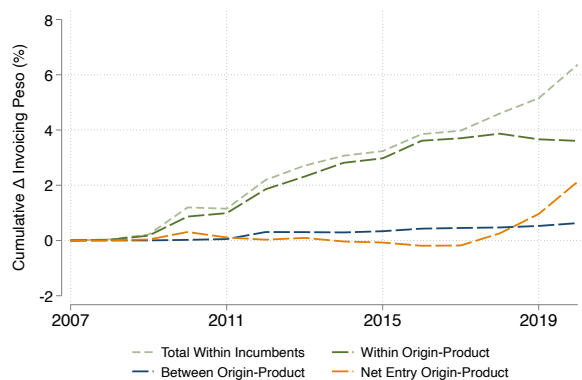
Figure 10: Decomposing Chilean Peso Invoicing Share over Time



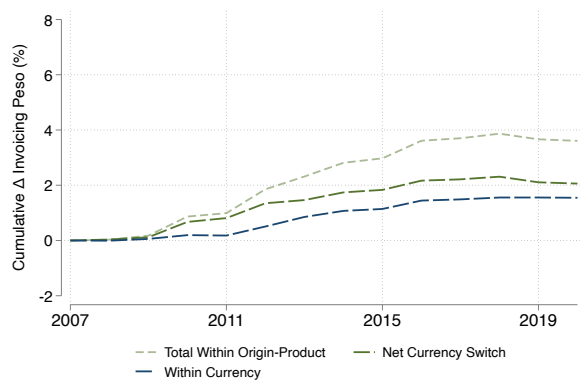
Panel (A): First Layer



Panel (B): First Layer



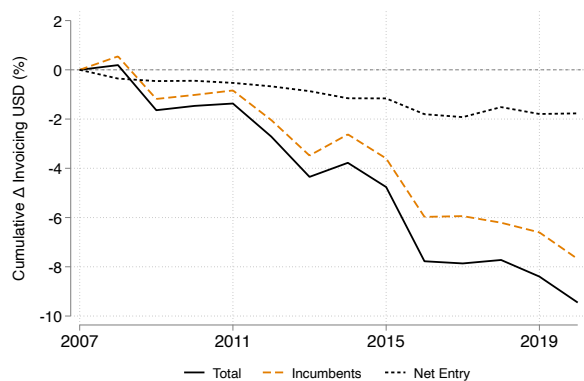
Panel (C): Second Layer



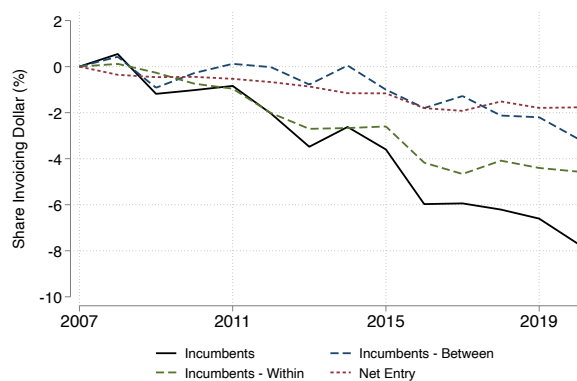
Panel (D): Third Layer

Notes: The Figure reports the results from the dynamic Olley–Pakes decomposition from Equations (1), (2), and (3) at the origin–product level for Chilean imports between 2007 and 2020. Each panel corresponds to a layer of the decomposition. We consider the share of imports invoiced in Chilean peso. Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year.

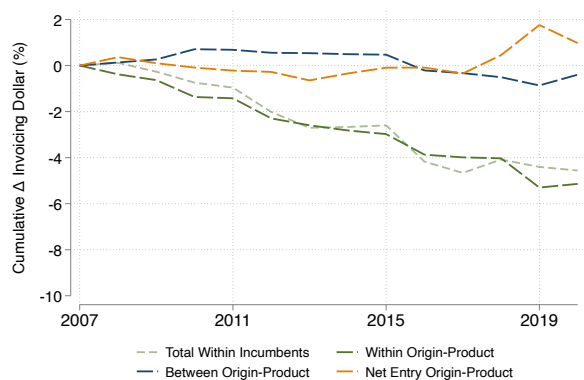
Figure 11: Decomposing US Dollar Invoicing Share over Time



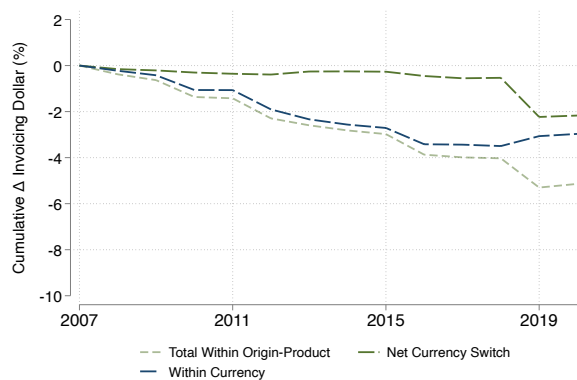
Panel (A): First Layer



Panel (B): First Layer



Panel (C): Second Layer

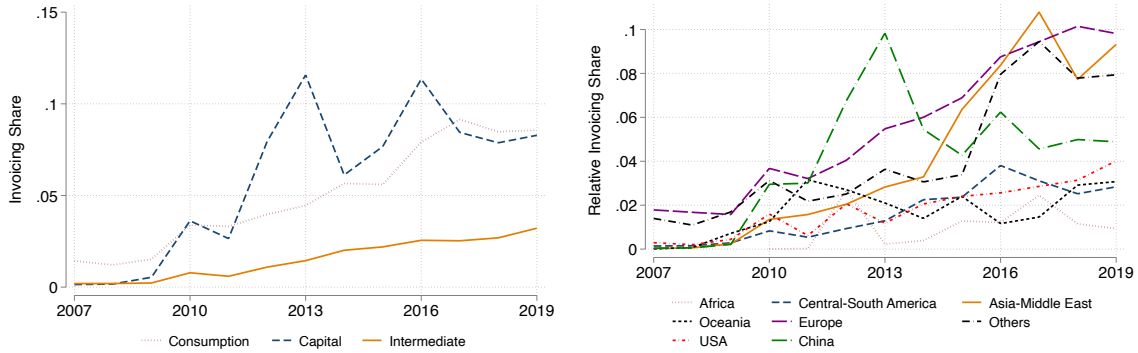


Panel (D): Third Layer

Notes: The Figure reports the results from the dynamic Olley–Pakes decomposition from Equations (1), (2), and (3) at the origin–product level for Chilean imports between 2007 and 2020. Each panel corresponds to a layer of the decomposition. We consider the share of imports invoiced in US dollar. Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year.

A.3 Additional Results: Product vs Origin

Figure 12: Subgroup by Origin or Product Type - Chilean Peso Invoicing



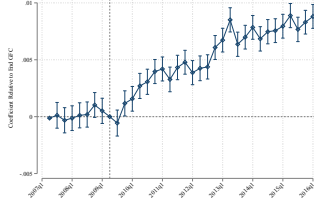
Notes: The left panel of the Figure shows the evolution of the invoicing share in Chilean Peso in imported goods within three broad product categories: Consumption, capital, and investment goods as defined according to the BEC classification. The right panel of the Figure shows the evolution of the invoicing share in Chilean Peso in imported goods within major trade partners: USA, Europe, China, Central-South America, Asia-Middle East, Africa and Others.

Table 12: Effect of GFC on Probability of Invoicing in Chilean Peso

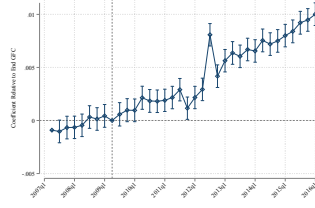
	Average Probability (Full Sample)		Average Probability (Switchers)	
	(1) Pr(PESO)	(2) Pr(PESO)	(3) Pr(PESO)	(4) Pr(PESO)
Post GFC	0.011*** (0.001)	0.012*** (0.001)	0.002*** (0.000)	0.002*** (0.000)
R^2	0.672	0.762	0.276	0.276
Firm $\times$ Product $\times$ Origin	YES	NO	NO	NO
Firm $\times$ Origin	NO	YES	YES	YES
Firm $\times$ Product	NO	YES	YES	YES
Observations	4009879	5004913	1996973	1996973

Notes: The Table reports estimates from the following linear probability model: $y_{imty} = \beta_1 \cdot \text{PostGFC}_t + \gamma_{im} + \epsilon_{imty}$. The dependent variable in Columns (1) and (2) is the probability of invoicing in Chilean peso, i.e. a dummy equal to one if the transaction at the imt level is invoiced in Chilean peso and zero if invoiced in US dollar. The main explanatory variable is a dummy for the post-GFC period (2009–2019). The dependent variable in Columns (3) and (4) is the probability of switching to Chilean peso invoicing, i.e. a dummy equal to one if the firm switches from invoicing in US dollar at $t - 1$ to invoicing in Chilean peso at t . In this case, we use single-currency importers. Standard errors are clustered at the im level. *** $p < 0.001$.

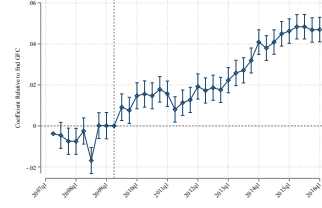
Figure 13: Effect of GFC on Probability of Invoicing in Chilean Peso: Main Origins



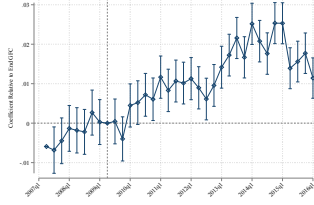
Panel (A): China



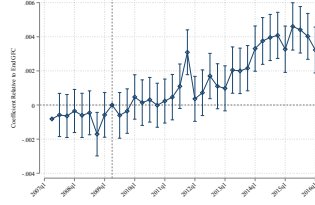
Panel (B): USA



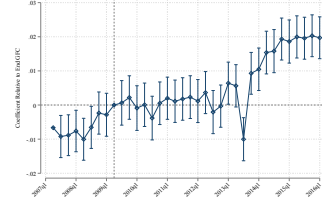
Panel (C): Germany



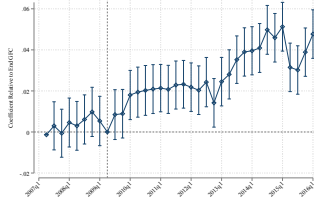
Panel (D): Not Specified



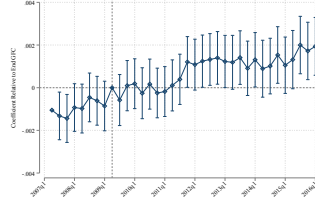
Panel (E): Brazil



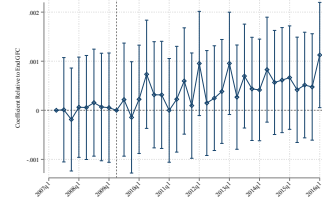
Panel (F): Italy



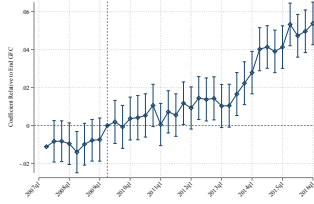
Panel (G): Spain



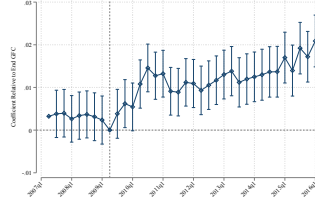
Panel (H): Argentina



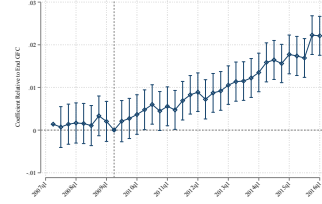
Panel (I): Perú



Panel (J): France



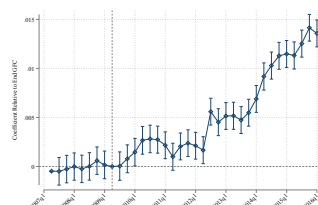
Panel (K): Japan



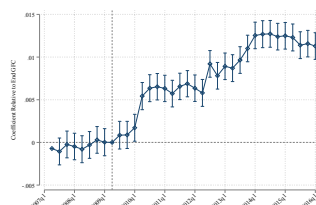
Panel (L): Mexico

Notes: The Figure reports the results from Equation (5) estimated separately for each of top twelve origin countries in the sample (ranked by number of observations at the *imt* level between 2007Q1 and 2019Q4). The dependent variable is the probability of invoicing in Chilean Peso, relative to 2009Q2 (the end of the GFC). The vertical dashed line corresponds to the NBER recession period for the end of the GFC. Standard errors are clustered at the importer–product–origin level.

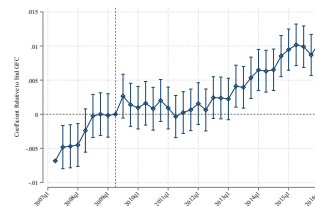
Figure 14: Effect of GFC on Probability of Invoicing in Chilean Peso: Main HS2



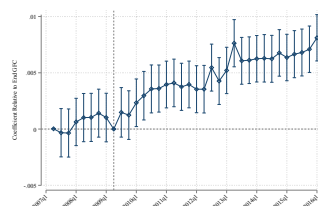
Panel (A): Machinery / Electrical



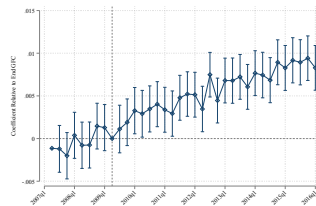
Panel (B): Machinery / Electrical



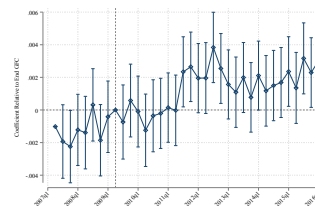
Panel (C): Optical, photographic, measuring, medical instruments



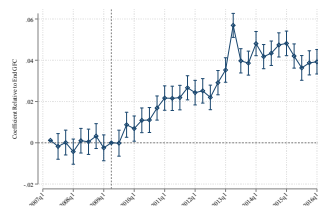
Panel (D): Plastics and rubber and articles thereof



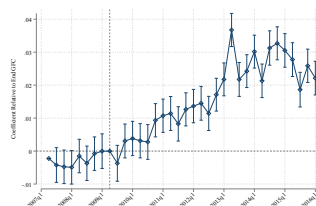
Panel (E): Articles of iron or steel (base metals)



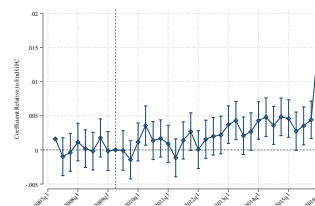
Panel (F): Vehicles (non-railway), parts accessories



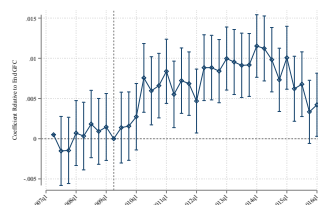
Panel (G): Textile apparel clothing accessories (knitted/crocheted)



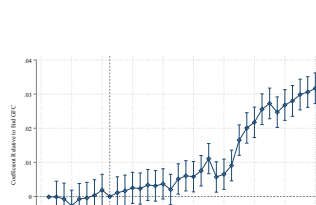
Panel (H): Textile apparel clothing accessories (other fabrics)



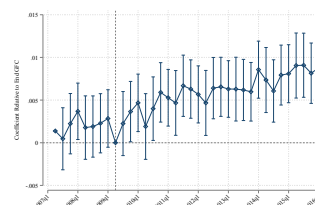
Panel (I): Tools, implements, cutlery, spoons forks (base metal)



Panel (J): Furniture; bedding; lighting fittings; prefabricated buildings



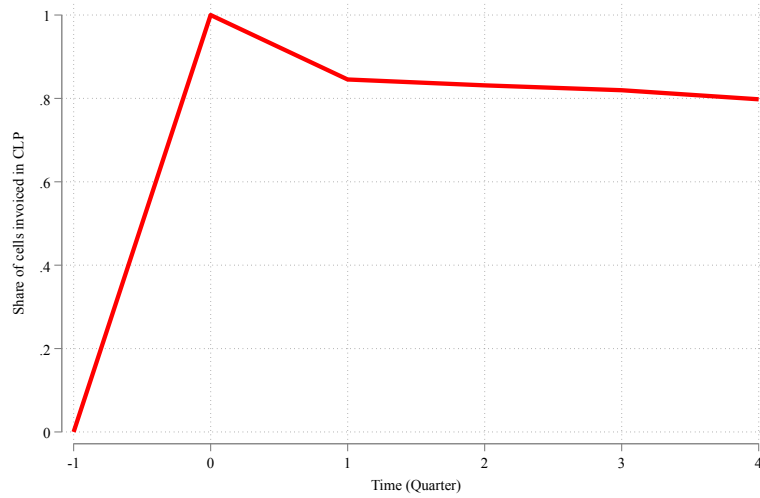
Panel (K): Plastics and rubber and articles thereof



Panel (L): Furniture; bedding; lighting fittings; prefabricated buildings

Notes: The Figure reports the results from Equation (5) estimated separately for each of top twelve HS2 products in the sample (ranked by number of observations at the *imt* level between 2007Q1 and 2019Q4). The dependent variable is the probability of invoicing in Chilean Peso, relative to 2009Q2 (the end of the GFC). The vertical dashed line corresponds to the NBER recession period for the end of the GFC. Standard errors are clustered at the importer-product-origin level.

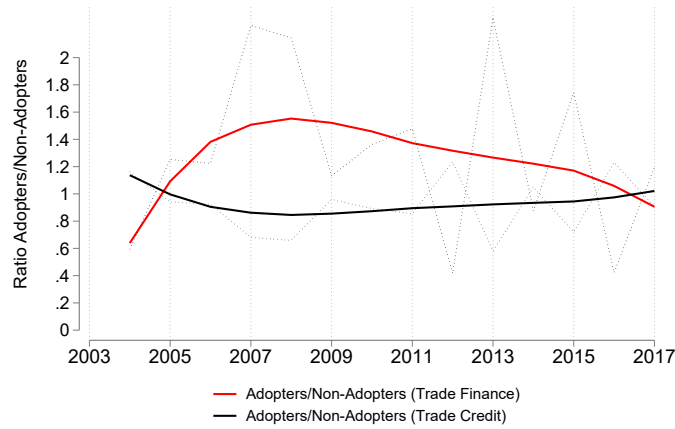
Figure 15: Persistence in Switching from US dollar to Chilean peso



Note: The Figure describes the invoicing currency in importer–origin-product im cells that switch for the first time from US dollars to Chilean peso in their imports to the four quarters after the switch. For each im cell, quarter zero corresponds to the first quarter in which the invoice currency is Chilean peso. We then compute the share of im cells invoicing in Chilean peso in each period.

A.4 Additional Results: Role of trade credit

Figure 16: Trade Credit and Invoicing



Notes: The Figure plots the ratio between the value imported by firms adopting of a new currency over the value imported by firms using US dollar invoicing. We compute the ratio distinguishing between transactions that took place under trade finance (i.e. pre-shipment payment) and under trade credit (i.e. post-shipment payment). Data are from [Garcia-Marin et al. \(2019\)](#).

A.5 Additional Results: Importers' Bargaining Power in Invoicing

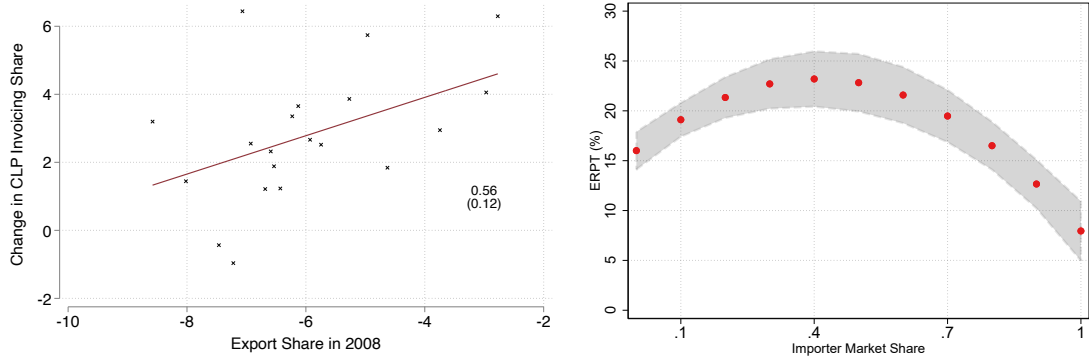
The invoicing pattern observed in the data implies that, at some point in time, exporters located in the origin countries started to accept payment in Chilean Peso and no longer, in the largely used, US Dollar. Arguably, the Chilean Peso is a less common currency compared to the ubiquitous US Dollar which thus suggests the existence of a relatively higher bargain power of Chilean importers when defining which currency to use to exchange goods internationally.

Therefore, we provide evidence that is coherent with the existence of a higher relative bargain power of Chilean importers with respect to foreign exporters.

First, the left panel of Figure 17 shows a positive linear relationship between the cumulative change in the aggregate share of Chilean Peso at the origin level and the Chilean export share in the origin's total exports in 2008. Intuitively, the graph suggests that, those origin countries whose total exports were largely dependent on Chile, they were also those countries whose share of transactions invoiced in local currency increased more. This is consistent with the idea that Chile was able to impose its currency to those counterparts that were more dependent from them.

Second, a model with strategic interactions in pricing among Chilean importers predicts an inverted U-shape relationship between exchange rate pass-through to import prices invoiced in local currency and importer's market share (Juarez (2024)). We confirm the existence of this pattern in the left panel of Figure 17.

Figure 17: Invoicing as Importers' Choice



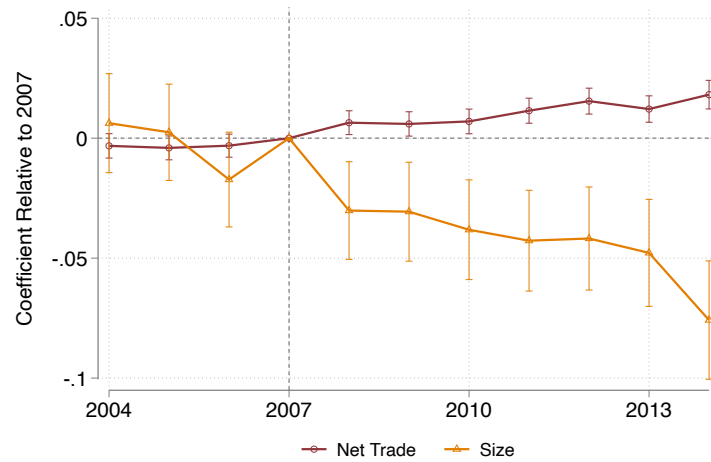
Notes: The left panel plots the relationship between the cumulative change in the aggregate share of CLP at the origin level and the Chilean export share in the origin's total exports in 2008 (in logs). We compute the cumulative change starting from 2008. We consider all horizons from 5 to 10 years, and plot the relationship after absorbing time fixed effects. The right panel plots the estimated quadratic relationship between the importer's market share and the pass-through rate of exchange rate fluctuations in import prices, estimated using the following specification:

$$\Delta \log p_{foit} = \beta_1 \Delta \log e_{ot} + \beta_2 S_{fsit} + \beta_3 \Delta \log e_{ot} \times S_{fsit} + \epsilon_{foit},$$

where S_{fst} is the market share of importer f in sector s (HS4-origin), and $\log p_{foit}$ is the unit import price in local currency for importer f , product i (HS8) from origin o .

A.6 Additional Results: Model Validation

Figure 18: Testing the Model - Event Study



Note: The Figure plots the coefficients β_τ from the event study specification in Equation (??), capturing the differential reduction in US Dollar invoicing by more exposed firms. Coefficient in 2007 is normalized to zero. Exposure is measured by: i) the difference between the total exports denominated in US Dollar and the total imports denominated in US Dollars normalized by the firm's total trade ("Net Trade"); ii) the market share of a firm in a given product-origin pair ("Size"). Data are from [Garcia-Marin et al. \(2019\)](#).

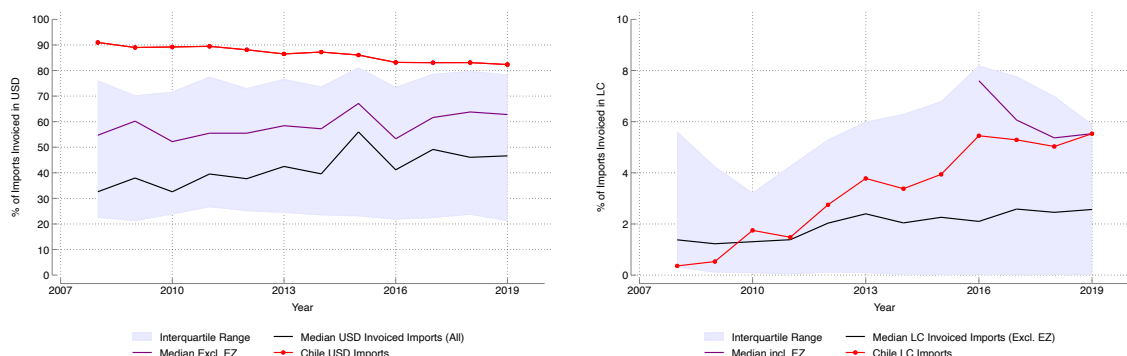
Table 13: Testing the Model - Sunk Cost

	Dummy USD				Share USD	
	(1) OLS	(2) OLS	(3) IV	(4) IV	(5) First Difference - IV	(6) First Difference - GMM
Lagged Dollar Share	0.160 (0.000)	0.062 (0.005)	0.061 (0.005)	0.158 (0.044)		
Lagged Δ Dollar Share					-0.382 (0.017)	-0.224 (0.013)
Controls	No	Yes	Yes	Yes	Yes	Yes
Origin $\times$ Year & Product $\times$ Year	Yes	Yes	Yes	No	No	No
Firm $\times$ Product $\times$ Origin & Year	No	No	No	Yes	Yes	No
Observations	1776386	768375	768375	623768	356394	260203

Note: The Table reports the coefficients from specification in Equation (19). The dependent variable in specifications (1) to (5) is a dummy variable equal to 1 when US Dollar invoicing is used and 0 otherwise; The dependent variable in specifications (5) to (7) is the (log) share of US Dollar invoicing. Column (6) is estimated in first difference via OLS. Column (7) is estimated using an Arellano-Bond GMM estimator using lagged levels of the right-side variables as instruments. "Lagged Dollar Share" refers to the lagged (log) share of US Dollar invoicing at the firm-origin-product level. Controls include: a natural hedging measure defined as the share of export invoiced in US Dollar; a strategic complementarities measure as the average share of imports invoiced in US Dollar among all competitors in a given sector(HS6)-origin pair. IV columns refers to the case in which the strategic complementarity measure is instrumented with the import-weighted average US Dollar exposure of all competitors in a given sector(HS6)-origin pair. Standard errors are clustered at firm level. We use data from official customs 2007-2019.

A.7 Additional Results: Cross-Country Evidence of Invoicing Currency

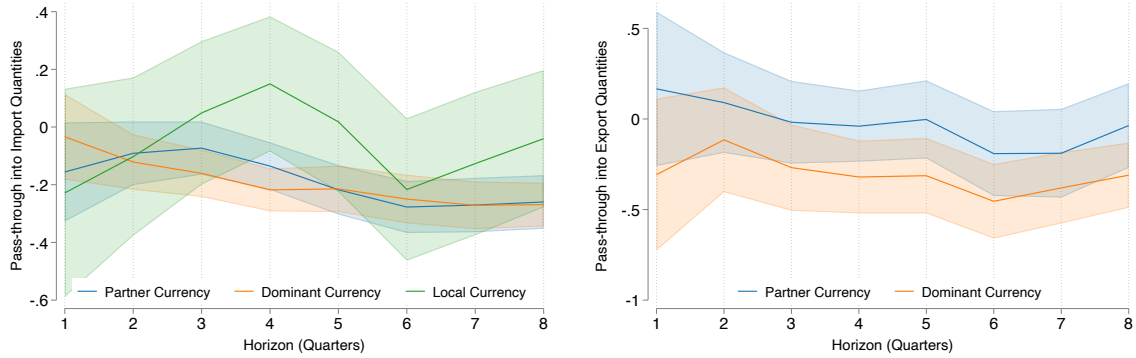
Figure 19: Cross-Country Evidence of Invoicing Currency



Notes: The figure uses data from [Boz et al. \(2020\)](#) which contains invoicing data for the period from 1990 to 2019 for 102 countries. The left panel reports the share of imports invoiced in US dollar. The red line shows the share of imports invoiced in US dollar for Chile, the black line reports the median share of imports invoiced in US dollar in the whole sample (including the Eurozone), the purple line reports the same statistics but without the Eurozone. The right panel reports the share of imports invoiced in local currency (LC). The median share of imports invoiced in local currency excluding the Eurozone (purple line) is reported from 2015 onwards only for readability purpose. The shaded area represents the interquartile range in the distribution of share of imports invoiced in each currency. The dataset is available at <https://www.imf.org/~media/Files/Publications/WP/2020/Datasets/wp20126.ashx>.

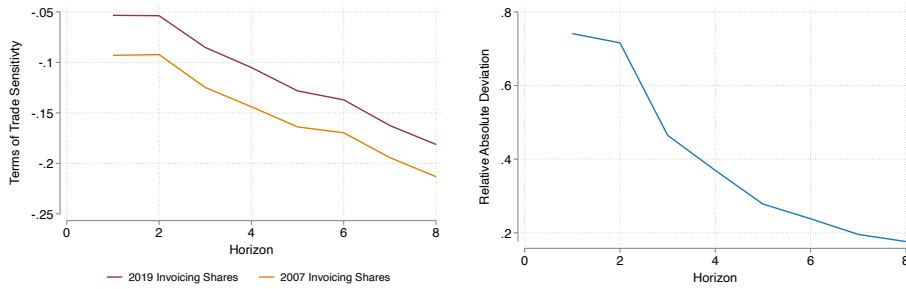
A.8 Additional Results: Counterfactual Exercises

Figure 20: Quantity Sensitivities to Exchange Rates



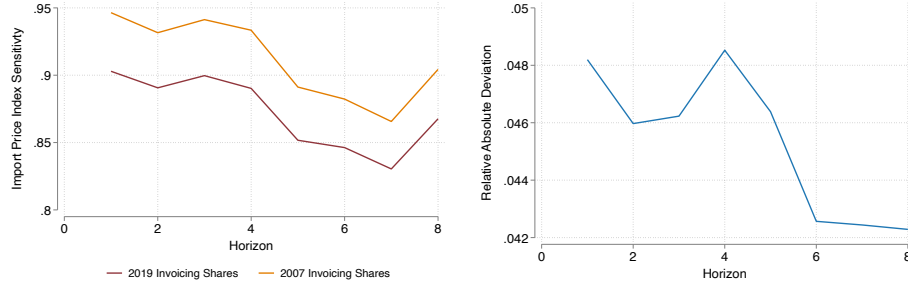
Notes: The left (right) panel show the estimated coefficients from the specification in Equation (20) using import (export) quantities as dependent variable. We use data from official customs running from 2007 to 2019 for imports and 2009 to 2019 for exports. We exclude the LCP case for export quantities because of the absence of sufficient observations in the estimation.

Figure 21: Terms of Trade Sensitivity to Exchange Rates



Notes: The left panel plots the 2-periods moving average of the response of the terms of trade (TOT) to a depreciation of the Chilean peso. For each horizon l , we use the following formula: $\Delta TOT_t^l = \Delta e \left[\sum_{\tau=0}^l \beta_{\tau}^{PX,CLP} \times S^{X,CLP} + \beta_{\tau}^{PX,D} \times S^{X,D} + \beta_{\tau}^{PX,p} \times S^{X,p} \right] - \Delta e \left[\sum_{\tau=0}^l \beta_{\tau}^{PM,CLP} \times S^{M,CLP} + \beta_{\tau}^{PM,D} \times S^{M,D} + \beta_{\tau}^{PM,p} \times S^{M,p} \right]$. We use the estimated coefficients from Equation (20). Invoicing shares are computed from official import (export) customs data from 2007 (2009) for the initial period (orange line) and from 2019 for the last period (red line). The right panel plots the difference between the response of the trade balance in 2007 and in 2019 relative to the 2019 case.

Figure 22: Terms of Import Inflation to Exchange Rates



Notes: The left panel plots the response of the import price index (IPI) to a depreciation of the Chilean peso. For each horizon l , we use the following formula: $\Delta IPI_t^l = \Delta e \left[\sum_{\tau=0}^l \beta_{\tau}^{PM,CLP} \times S^{M,CLP} + \beta_{\tau}^{PM,D} \times S^{M,D} + \beta_{\tau}^{PM,p} \times S^{M,p} \right]$. We use the estimated coefficients from Equation (20). Invoicing shares are computed from official import customs data from 2007 for the initial period (orange line) and from 2019 for the last period (red line). The right panel plots the difference between the response of the trade balance in 2007 and in 2019 relative to the 2019 case.

B Appendix: Theory

B.1 Firm Invoicing Choice.

Start from the firm maximization problem displayed in Equation (10). For any given η_i , the optimal domestic price $p_i^{d*} = \frac{\theta}{\theta-1} E[C_i(s^v, \eta_i)]$. A firm invoices working capital x_i in vehicle currency v ($\eta_i^* = 0$) if and only if expected profit for doing so is higher or equal to the expected profit in the complementary case when working capital is invoiced in domestic currency ($\eta_i^* = 1$). That is:

$$\eta_i^* = 0 \quad \text{iff} \\ p_i^d (p_i^d)^{-\theta} - \mathbb{E}[C_i(s^v, 0)] (p_i^d)^{-\theta} - F_i^v \geq p_i^d (p_i^d)^{-\theta} - \mathbb{E}[C_i(s^v, 1)] (p_i^d)^{-\theta} - F_i^d$$

Plug in the optimal price expression for the respective cases and obtain:

$$\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 0)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 0)] \right)^{-\theta} - \mathbb{E}[C_i(s^v, 0)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 0)] \right)^{-\theta} - F_i^v \geq \\ \frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 1)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 1)] \right)^{-\theta} - \mathbb{E}[C_i(s^v, 1)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 1)] \right)^{-\theta} - F_i^d$$

Which simplifies to:

$$\frac{(\theta-1)^{\theta-1}}{\theta^\theta} [\mathbb{E}[C_i(s^v, 0)]^{1-\theta} - \mathbb{E}[C_i(s^v, 1)]^{1-\theta}] \geq F_i^v - F_i^d$$

Then, consider the expression for the expected marginal cost of the firm for the two different cases:

$$\mathbb{E}[C_i(s^v, 0)] = w^{1-\alpha_i} \mathbb{E}[(1+i_v)\rho_v s_v]^{\alpha_i} \\ \mathbb{E}[C_i(s^v, 1)] = w^{1-\alpha_i} [(1+i_d)\rho_d]^{\alpha_i}$$

Re-write the expression above as:

$$\mathbb{E}[(1+i_v)\rho_v s_v]^{\alpha_i(1-\theta)} - [(1+i_d)\rho_d]^{\alpha_i(1-\theta)} \geq \tau_i \times (F_i^v - F_i^d) \\ \tau_i \equiv \frac{\theta^\theta}{(\theta-1)^{\theta-1}} \times \frac{1}{w^{(1-\alpha_i)(1-\theta)}}$$

which is the expression in the main text. Note that $\frac{\theta^\theta}{(\theta-1)^{\theta-1}} > 1$ for $\theta > 1$ and it is strictly increasing in θ .

Then, consider the simplified case in which $F_i^v = F_i^d$ and $s^v \sim \text{Log-Normal}(\mu, \Sigma)$. Then,

$$((1+i_v)\rho_v)^{\alpha_i(1-\theta)} \mathbb{E}[s_v^{\alpha_i}]^{1-\theta} \geq [(1+i_d)\rho_d]^{\alpha_i(1-\theta)}$$

Then, i) elevate both sides to the power of $1/(1-\theta)$, ii) take logs on both sides, iii) use the

result that $\mathbb{E}[s_v^{\alpha_i}] = \exp(\alpha_i \mu + 0.5 \times \alpha_i^2 \Sigma)$, and obtain the expression in the text.